Roadmap for Lectures 6-9

- Efficiency of competitive market equilibria; CS, PS and Welfare
- Competitive markets: Taxi-cab example (part 1)
- Competitive markets: Taxes, subsidies and externalities
- Competitive markets: Price controls
- Competitive markets: International trade with quotas and tariffs
- Monopoly - basic model
- Advanced applications of monopoly model
- Introduction to game theory
- Oligopoly - Cournot's output game
- Capturing consumer surplus
  - Direct price discrimination
  - Indirect price discrimination (nonlinear pricing, quality pricing)

Day 6:
Applications in competitive markets - taxicabs, taxes, subsidies

Topic 1: Market equilibria and efficiency

- The main new insight from all of our analysis of supply and demand is their marginal property:
  - Buyer demand curves indicate the marginal value of each unit of consumption.
  - Firm supply curves indicate the marginal cost of each unit of production.

    You may be wondering about this last statement since supply curves are only marginal cost curves for p greater than the relevant minimum average (opportunity) cost. That said, it is correct to think about the minimal amount of production, the MES, as a lumpy marginal cost for a possibly large MES batch of output. In this sense, the supply curve is the relevant marginal cost for production.

Topic 1: Market equilibria and efficiency

- **Market demand:** Each buyer's demand curve characterizes the various marginal values of consumption. Because the market demand curve is the horizontal sum of every buyer's demand curve, the market demand curve also has the marginal value property as well.
  - For any consumption level, determine the associated market price that would sell exactly those units. This price is the marginal value of the marginal buyer of that unit.
  - Thus, market demand curves indicate the marginal value of each unit of consumption.

Topic 1: Market equilibria and efficiency

- **Market supply:** Each firm's supply curve characterizes the various "marginal" costs of production. Because the market supply curve is the horizontal sum of all the relevant firms' supply curves, the market supply curve also inherits this marginal cost property.
  - For any production level, determine the associated market price that would induce the production of exactly that number of units. This price is the marginal cost of the marginal producer of the marginal unit.
  - Thus, market supply curves indicate the marginal cost of each unit of production.

Topic 1: Equilibria, efficiency, continued

- What do the marginal properties of demand and supply imply for market equilibrium?:
  - **Buyer efficiency**. Every buyer who values the product more than the marginal cost of the market producing one more unit will consume the product.
  - **Seller efficiency.** Every seller who can produce a unit of output for less than marginal buyer's value does so.
  - **Aggregate efficiency.** The outcome is efficient in the sense that no other allocation of production and consumption can be found for which somebody gains and nobody loses. (Pareto efficiency.)
    - Note if we are not at the market equilibrium, we can generally change the allocation, making some people better off without harming others. Why?

Social (or aggregate) Surplus

- **Competitive markets are efficient:** there is no reallocation of production and consumption that can improve an actor's position without harming another.
- **Consumer surplus:** Recall that "consumer surplus" is the area under a buyer's demand curve and above the market price. The aggregate (or market) consumer surplus is the area between the market demand curve and the price.
  - Because market demand is the horizontal sum of the individual demands, the market consumer surplus is equal to the sum of the individual consumer surpluses.
- **Producer surplus:** Analogous to consumer surplus, producer surplus (or economic profit) is the area between the market price and a firm's supply curve. Aggregate or market producer surplus is the area between the market price and the market supply curve.
  - NB: this last statement must be modified if input costs are not constant with industry scale; we will do this below.

Producer surplus (a.k.a. economic profit)

- Key result: Producer surplus is equal to the area between the market price and the firm's supply curve. Why?
  - The revenue from selling q is the rectangle defined by the market price and the output supplied (which is where price intersects the marginal cost curve).
  - The opportunity cost of producing q is the area under the supply curve from the origin to *q*. To see this ...
    - Let $\hat{q}$ be the firm's minimum efficient scale (which may be zero if there is no fixed and avoidable cost of production).
    - The cost of producing the first $\hat{q}$ units is the area under the supply curve from 0 to $\hat{q}$ (which is a rectangle defined by the origin and the minimum average opportunity cost):

$$C(\hat{q}) = AC_{min} \cdot \hat{q}$$

    - The cost of (q - $\hat{q}$) additional units is the integral of the marginal cost curve (the area underneath the *MC* curve) from $\hat{q}$ to *q*.
    - Put these two pieces together -- the opportunity cost of producing *q* units is the area under the firm's supply curve from 0 to *q*.

Producer surplus (a.k.a. economic profit)

- Because economic profit is revenue minus opportunity cost, the difference between the revenue rectangle and the area under the firm's supply curve is economic profit.
- This difference in area is exactly the area between the market price (the upper edge of the revenue rectangle) and the firm's supply curve.
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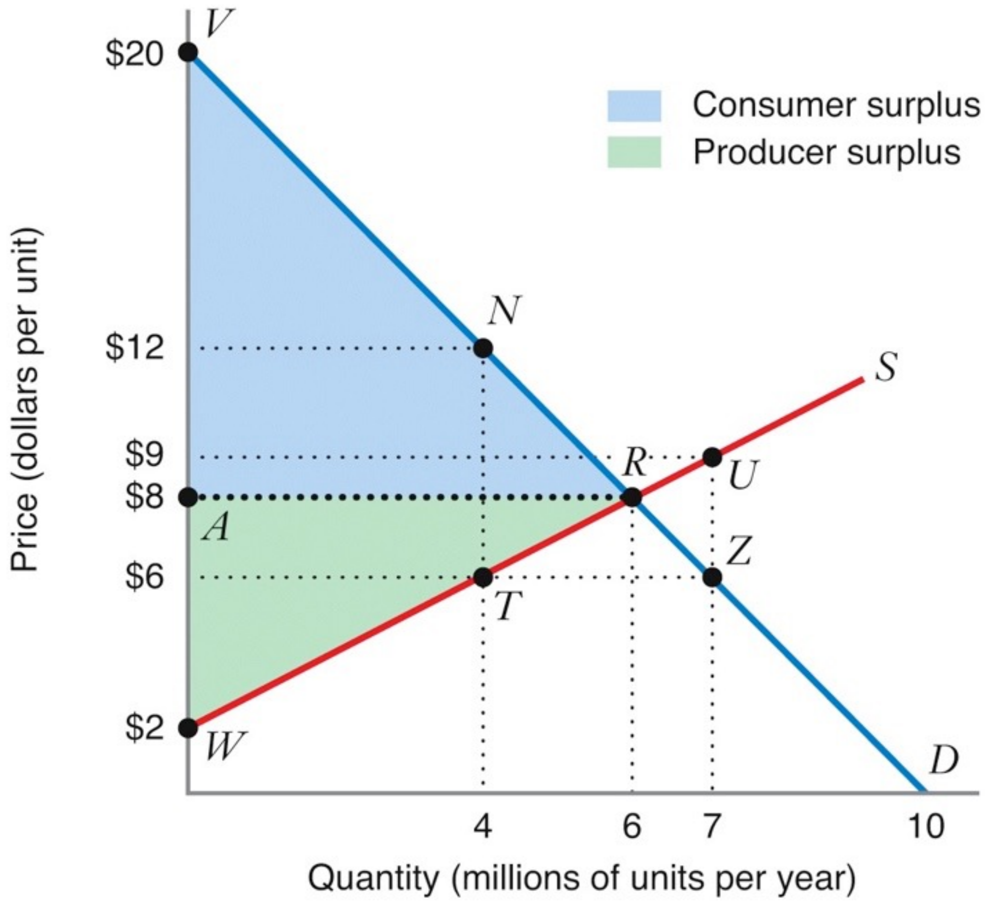
Aggregate Consumer and Producer Surplus



Social (or aggregate) Surplus

- Social or Aggregate Surplus in the market is the sum of the market's consumer and producer surpluses.
- Another way of saying that the competitive outcome is efficient is that the aggregate surplus is maximized.
- No change in the amount of output produced, how this output is allocated across the various suppliers, or how the consumption is allocated across the various buyers can increase aggregate surplus.
- Adam Smith's The Wealth of Nations (1776) described the "invisible hand" of the market and noted that the self-interested actions of each individual lead to such economic efficiency:
  - "He intends only his own gain, and he is in this…led by an invisible hand to promote an end which was no part of his intention."
- When a departure from perfect competition reduces the social surplus, we all the reduction deadweight loss.

Working example: Taxicabs

- We will make use of the following toy model of a taxicab market in a small town. It is unrealistic, but still insightful.
- The inverse demand function is

  $$P(Q) = 1 - Q/(100{,}000).$$

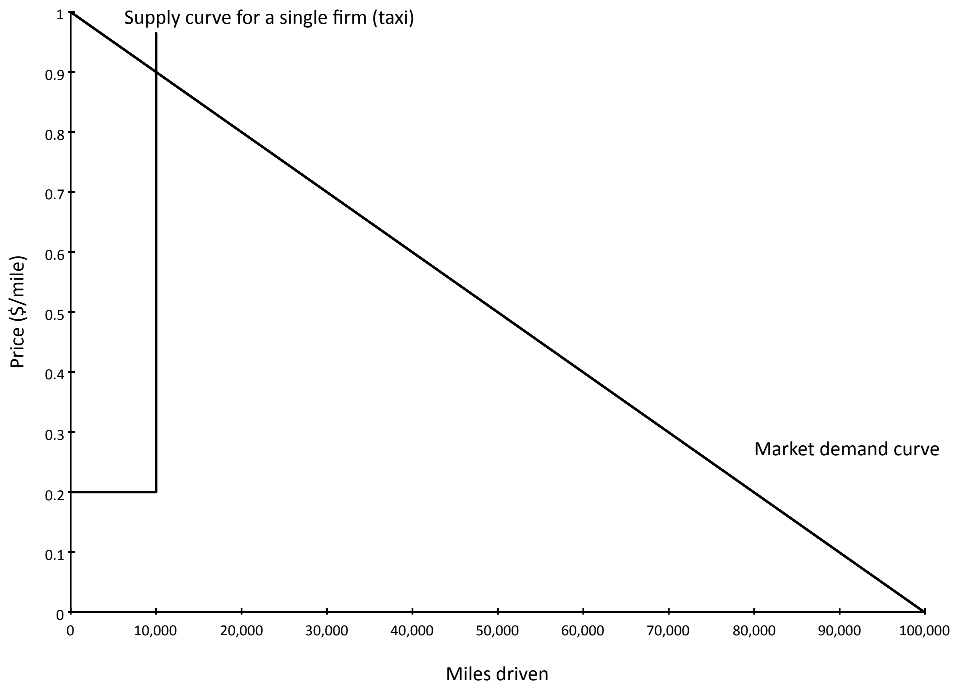
  Q is miles demanded per period; price is dollars/mile.
- Assume that each taxicab firm can supply up to 10,000 miles per period, its capacity. The opportunity cost for $q \leq 10{,}000$ is

  $$C(q) = 0.20\ q.$$

  This cost includes all inputs (fuel, labor, insurance, capital costs, depreciation/service for car, etc.). For simplicity, I have assumed there are no fixed costs.
- There are many (i.e., more than 8) potential taxicab driver-firms and they are all price takers.

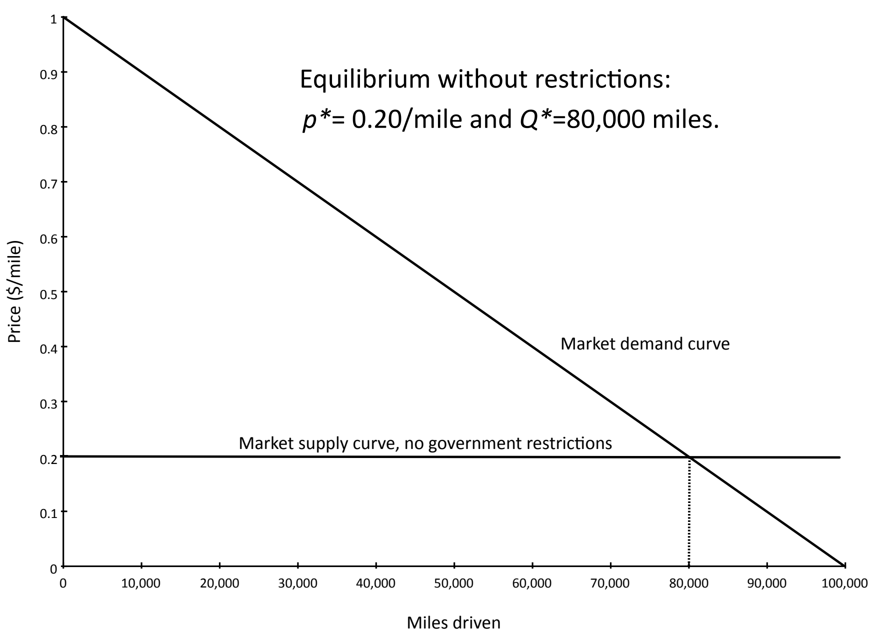
Working example: Taxicabs

- Graph of market demand and firm supply curves:



Working example: Taxicabs

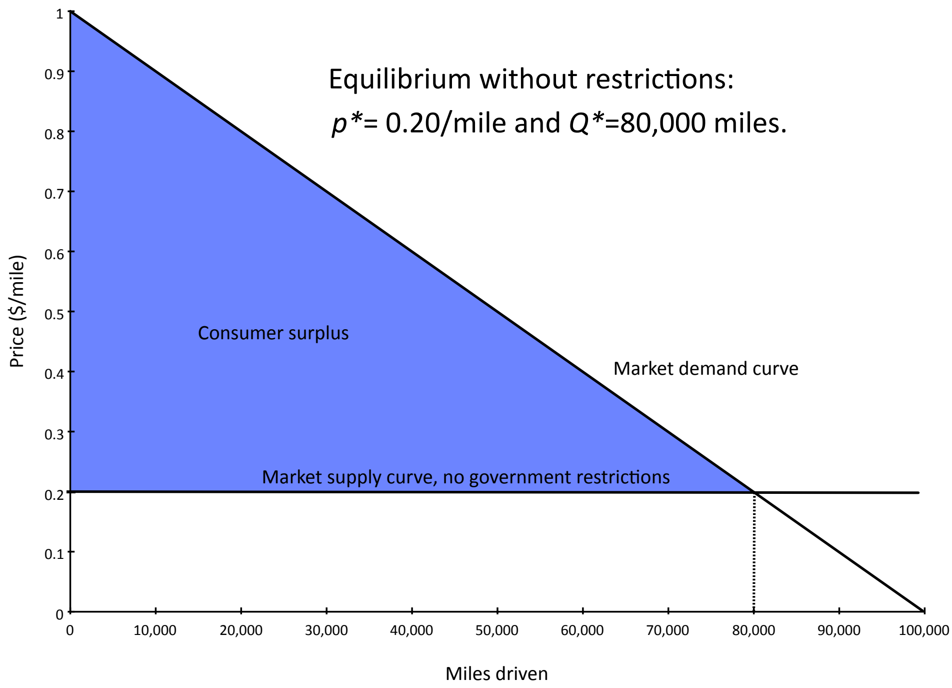
- Market equilibrium without restrictions on entry. Efficient.



- Firm supply unlimited miles at p* = 0.20/mile. Q*= 80,000.
- Each firm earns zero economic profit.

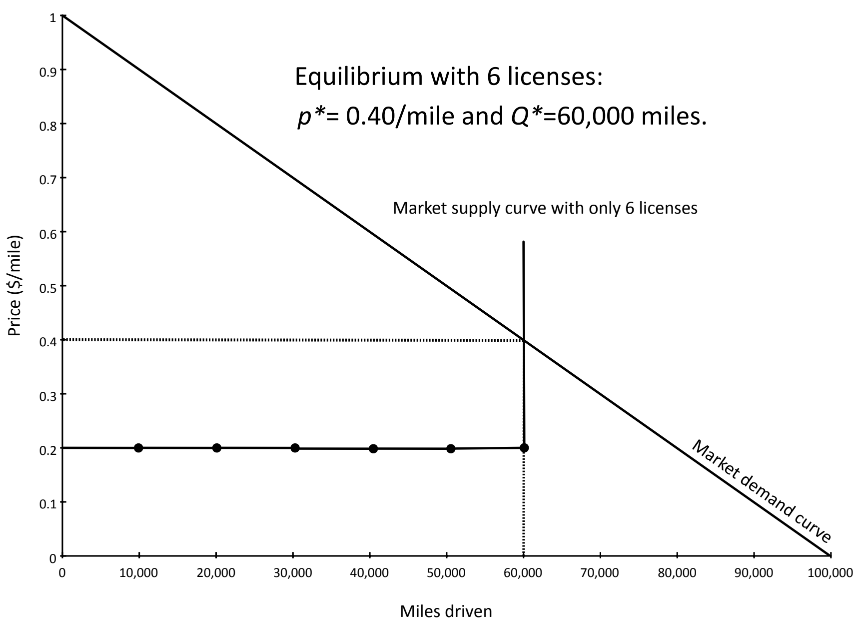
Working example: Taxicabs

- All surplus is consumer surplus. Trade is efficient.



Working example: Taxicabs

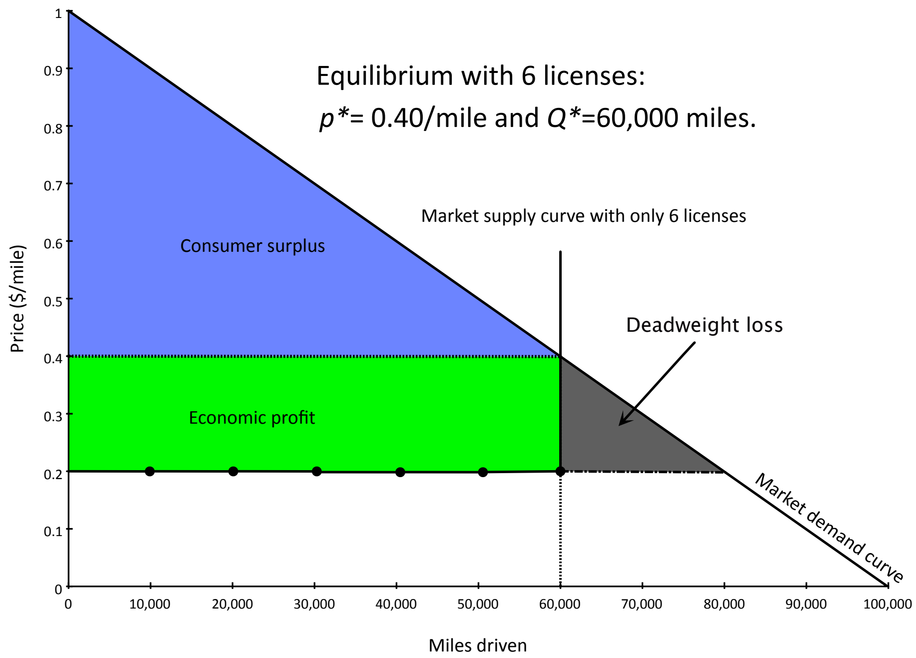
- Now suppose the government grants 6 non-transferable licenses to drive taxicabs. No one else may be in the industry.



- Now there are only $Q^* = 60{,}000$ miles supplied and the price is higher at $p^* = 0.40$/mile.

Working example: Taxicabs

- With entry restrictions, the 6 lucky licensees each earn economic profit of $\pi = \$2000$/year. Aggregate surplus is lower, however.



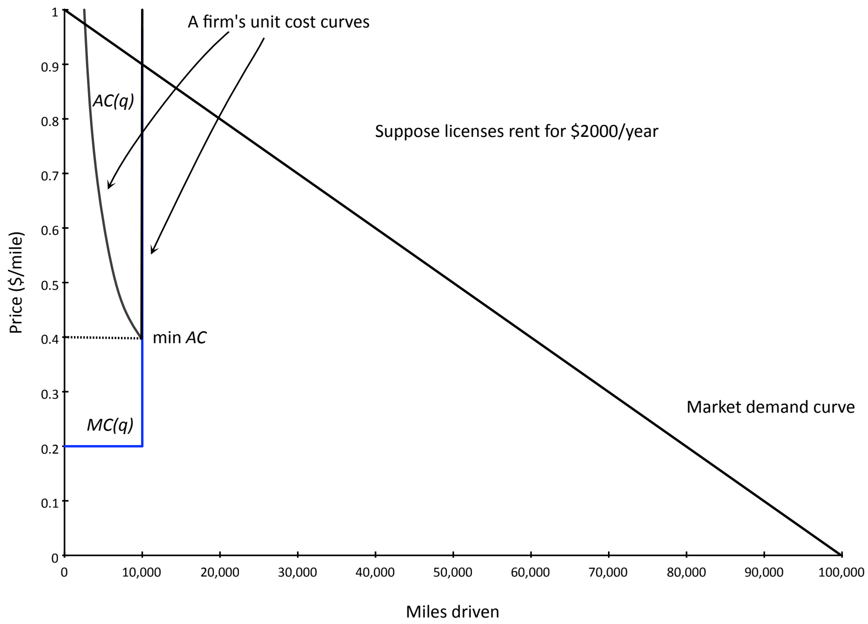
- The deadweight loss of the entry restriction is the reduction in aggregate surplus caused by 20,000 fewer miles supplied.

Remarks on deadweight losses

- Note that the deadweight loss in this example lies in between the efficient output of *Q*=80,000* and the actual output of *Q*=60,000*. This will generalize to other applications.
- There is a story behind the loss in surplus.
  - Take a buyer on the demand curve between these outputs and consider an unlicensed seller. This buyer values the output at a margin greater than the cost to this seller but less than the market price of *p*=0.40*. As such, there are gains from trade that the license restriction has prohibited. These lost trades add up to the deadweight loss in this example.
- Similar notions of deadweight loss arise in other applications.
  - deadweight loss will be in a region between the efficient and actual levels of production,
  - the source of the deadweight loss can be thought of as surplus-generating trades which the restriction prohibits or surplus-lowering trades that the restriction creates.

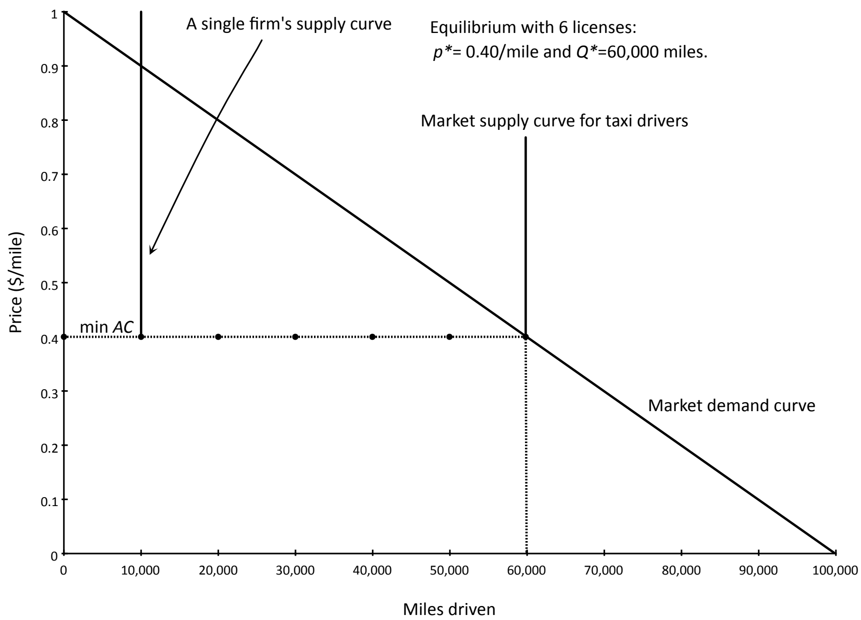
Working example: Taxicabs

- Now suppose that there are still only 6 licenses, but that license owners may rent their license if they don't use it themselves.
- If license rentals were free, economic profit would be $2000/year. Thus, we expect the rental price to be exactly $2000/year.



Working example: Taxicabs

- What will market supply look like when licenses can be rented?



- Note: no taxi-firm earns economic profit! The economic profits appear as a once-off capital gain to the initial 6 license holders.

Working example: Taxicabs

- Economic profits accrue only to the scarce factor. In our present taxicab setting, the only scarcity is the license. Driving a taxicab generates no profit. Owning the license creates an initial capital gain equal to the present value of a $2000/year revenue stream.
  - **Note**: A taxicab driver that is also a license owner still makes zero economic profit driving a taxicab. Why? Because the opportunity cost of using the license is $2000/year.
- Now let's add some additional complexity. Suppose that there is a single driver that is able to drive 20,000 miles per period. You can think of this scarce advantage as a better technology for extracting value from a license.
  - In equilibrium, we expect the high-capacity driver to rent a license and drive a taxicab. Furthermore, this driver will also earn some economic profit from his competitive advantage. Of course, if many drivers could also drive 20,000 miles, this would no longer be true.

Taxicabs with driver differences

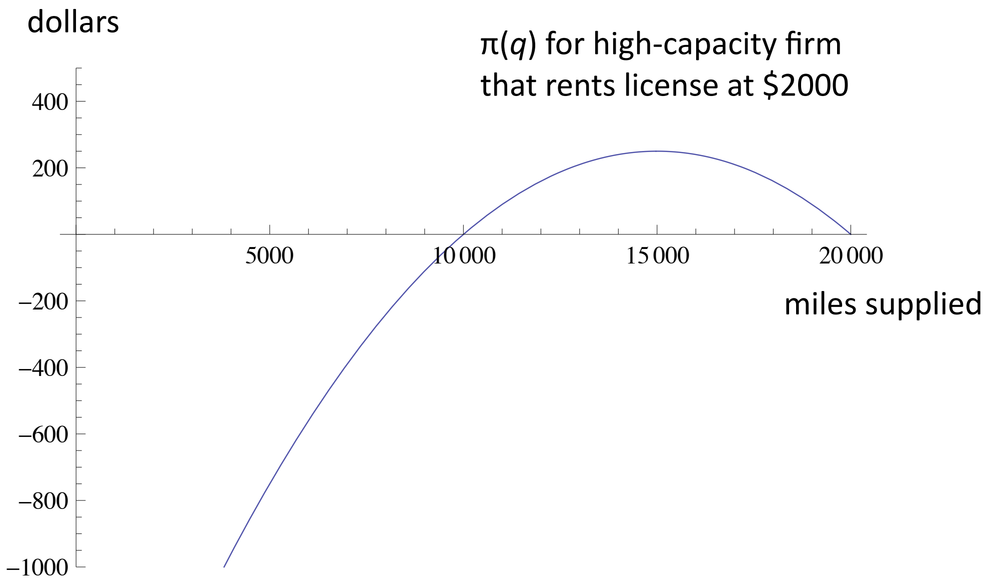
- Let's think more about what we expect happens with the introduction of a single high-capacity driver. In particular, are we so sure that the high capacity driver will produce at capacity?
  - If the high-capacity firm remains a price taker ignorant of his impact on the market price, then he will produce at capacity, q=20,000. Total supply will be Q=70,000 and the market price will be p=0.30. His profits in this case are

    $$\pi = (p - 0.20)(20{,}000 \text{ miles}) - \text{rent for license} = \$2000 - \text{rent}.$$

    Remarkably, he would make the same as choosing *q=10,000* and generating a market price of *p=0.40*.
  - In this case, it seems implausible that the high-capacity driver would remain a price taker. When we develop the model of monopoly, we will see that this driver does best by producing below capacity at *q=15,000* at an equilibrium price of *p=0.35*. We will also see that the low-capacity drivers will still prefer to produce at their capacities of 10,000 each.

Taxicabs with driver differences

- It is a coincidence that *q=10,000* and *q=20,000* generate the same economic profit of zero. The highest economic profit can be attained at *q=15,000*



Taxes, subsidies, externalities

- Specific (or "unit") taxes under perfect competition
  - irrelevance of who has legally obligation to pay the tax
  - economic incidence depends upon S & D elasticities
  - deadweight loss (or "excess burden"); too little trade
- What about valued-added taxes (proportional to price)?
- General versus partial equilibrium tax analysis?
- Subsidies are the mirror image of taxes
  - economic incidence of subsidy depends upon elasticities
  - deadweight loss from too much trade
- Can taxes or subsidies ever be efficient?
  - externalities generate a positive role for taxes or subsidies
  - two different approaches: Pigou and Coase

Taxes, subsidies, externalities - continued

- What do the marginal properties of supply and demand curves tell us about taxes?
  - A specific or per-unit tax on a seller of a fixed amount per unit sold (regardless of price) implicitly shifts the firm's marginal and average cost curves up by the amount of the tax. This, in turn, shifts supply curves by the same amount.
  - A specific or unit tax on buyers has the reverse implication: demand curves shift downward by the amount of the unit tax.
  - Specific or per-unit subsidies are like negative taxes. Easy to understand once you understand the impact of taxes.
  - A value-added (ad valorem) or proportional tax is a bit more complicated because it rotates supply and demand curves depending on the market price. More on this later.

More definitions

- The legal incidence of the tax is on the seller if the seller is legally required to pay the tax for each unit sold; the legal incidence is on the buyer if the buyer is required to pay the tax for each unit purchased.
  - e.g., the legal incidence of sales taxes in the US and VAT/GST taxes in Europe for hotels, is typically on the buyer; the tax is not included in the seller's quoted price and the buyer is obliged to pay the extra amount.
  - the legal incidence of most VAT/GST taxes in Europe (excepting hotels) falls on the seller; the seller's offered price already includes the required tax and the buyer is not obliged to pay an additional amount.
- The economic incidence of a tax is the net amount of the tax that is paid by each side after accounting for tax-induced changes in price. This is the interesting notion of incidence.
  - E.g., if the seller is legally required to pay a tax of $100/unit sold but the price increases by $75, then the economic incidence on the seller is only $25 while the economic incidence on the buyer is $75.

Some additional notation for prices and tax

- To be clear about economic incidences of a tax, we need to introduce additional prices:
  - p is the pre-tax market price;
  - t is the per-unit tax;
  - p_b is the buyer's tax-inclusive payment (includes any tax obligations);
  - p_s is the seller's tax-inclusive revenue (includes any tax obligations).
- If the legal incidence is on the seller, then $p_b = p, \; p_s = p - t$.
- If the legal incidence is on the buyer, then $p_b = p + t, \; p_s = p$.
- Regardless of the legal incidence, it must be that

$$p_b - p_s = t.$$

  The difference between what a buyer pays for a unit and what a seller receives for the same unit is the amount of unit tax paid.

Measuring economic incidence

- Suppose that the equilibrium market price before any tax is imposed is p_0^*.
- Once the tax is imposed, denote the equilibrium market after-tax prices for the buyer and seller as p_b^* and p_s^*, respectively.
- The economic incidence for the buyer is measured by

$$\Delta p_b = p_b^* - p_0^*$$

  and the economic incidence for the seller is measured by

$$\Delta p_s = p_0^* - p_s^*.$$

  Of course, the total economic incidence equals the per-unit tax collected:

$$\Delta p_b + \Delta p_s = t.$$

Behavioral assumption

- We make a simple behavioral assumption that seems reasonable, but perhaps not always true: demand depends only upon p_b and supply depends only upon p_s. Only after-tax prices are relevant.
  - This is not always the case. I expect that hotels do not include VAT in their listed rates because only infrequent (non-repeat) customers would not know to account for the extra tax at the time of booking. Such customers may be surprised to find that they owe more than they budgeted, but will probably pay their bill. In this case, demand depended on p, not p_b.
  - Some grocery store experiments also confirm that variations in p but not p_b will affect demand, even in cases where the consumers probably know both p and p_b. I suspect that with repeat purchases and binding budget constraints, consumers eventually turn their attention exclusively to p_b.

Taxes, subsidies, externalities - continued

- Given our behavioral assumption, we have a powerful conclusion about tax incidence in competitive markets:
  - the economic incidence of a tax is entirely determined by the market supply and demand curves; legal incidence is irrelevant.
- There is a simple mathematical proof of this result using our new notation for prices and the tax. There is also a graphical “proof” of the result that is more illuminating about what really matters for economic incidence. We will look at it both ways.

The mathematics of economic incidence

- Given supply and demand functions which depend only on after-tax prices (our behavioral assumption), market equilibrium requires

$$D(p_b)=S(p_s).$$

- Given the per unit tax is t, by definition we must also have

$$p_b - p_s = t.$$

- This gives us two equations and two unknowns. With very mild restrictions on the supply and demand curves, we will find that there is a single pair of after-tax prices that solve both equations. The after-tax prices do not depend on the legal tax incidence.
- Because economic incidence depends only upon the original no-tax, p_0^*, and the new after-tax equilibrium prices p_b^* and p_s^*, it follows that economic incidence does not depend upon the legal incidence of the tax.

The mathematics of economic incidence

- We can calculate the economic incidence by knowing only the supply and demand functions and the amount of the unit tax t.
- The original no-tax equilibrium price p_0^* satisfies

$$D(p_0^*)=S(p_0^*).$$

- After the tax is imposed, using the fact that $p_b - p_s = t$, the after-tax buyer's price satisfies

$$D(p_b^*)=S(p_b^*-t).$$

and the after-tax seller's price satisfies

$$D(p_s^*+t)=S(p_s^*).$$

- Immediately we have the conclusion that $p_b^* \geq p_0^* \geq p_s^*$ and the gaps between these prices are determined by the slopes of the supply and demand equations near the equilibrium point.

The mathematics of economic incidence

- As an extreme example, suppose that the supply curve is perfectly inelastic (vertical) at $Q^S=100$.
- Then p_0^* must satisfy

$$D(p_0^*)=100.$$

- The after-tax buyer's price must also satisfy

$$D(p_b^*)=100.$$

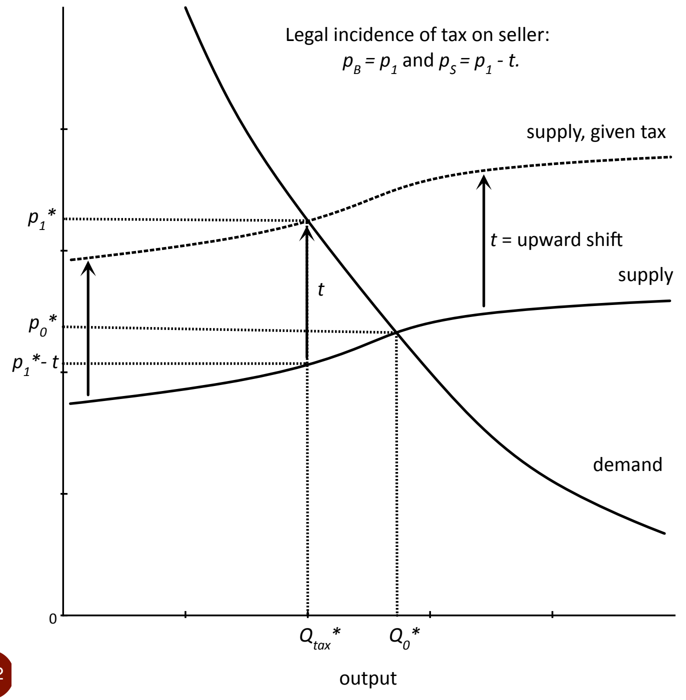
- It follows that $p_b^* = p_0^*$, and so $p_s^* = p_0^* - t$. The entire economic incidence falls on the sellers and the buyers are unaffected.
- Equilibrium requires that the buyers continue to purchase the inelastically supplied 100 units, so their after-tax price cannot change. If some of the tax was passed onto the buyers, their demand would fall below 100 and demand would be below supply at that price. Anyway you look at it, all of the change must be into the sellers' after-tax price to maintain equilibrium.

The mathematics of economic incidence

- If instead of inelastic supply, we had perfectly inelastic demand, the reverse would arise. All of the tax would be passed along to the inelastic side of the market (buyers), but their demand would not change. The supply side would not bear any of the tax in this case and economic incidence entirely falls on the demand side.
- Suppose instead that supply is upward sloping but demand is perfectly elastic. If any part of the tax is passed onto buyers, demand will fall to zero. Thus, the entire tax must fall on the sellers (just as in the case in which supply was perfectly inelastic).
- Similarly, when supply is perfectly elastic, the economic incidence of the tax must fall entirely on the demand side.
- The theme that appears is that the more inelastic side of the market bears more of the economic tax incidence.
- To develop this theme generally, we turn to a graphical approach.

The geometry of economic incidence

- Suppose that the tax is legally imposed on sellers. It is as if each seller's unit cost functions shift upwards by t.

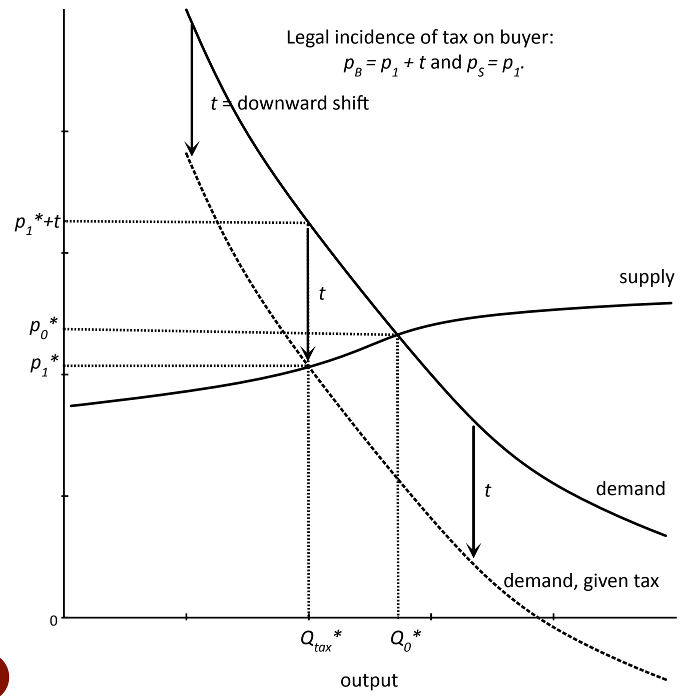


The new equilibrium price is p_1^* which is also the net amount the buyer pays, $p_1^* = p_b^*$. The seller receives net after tax, $p_s^* = p_1^* - t$.

In this example, the buyer the larger economic incidence of the tax.

Geometry of econ incidence, continued

- Suppose instead that the tax is legally imposed on buyers. It is as if each buyer's demand curve shifts downwards by t.



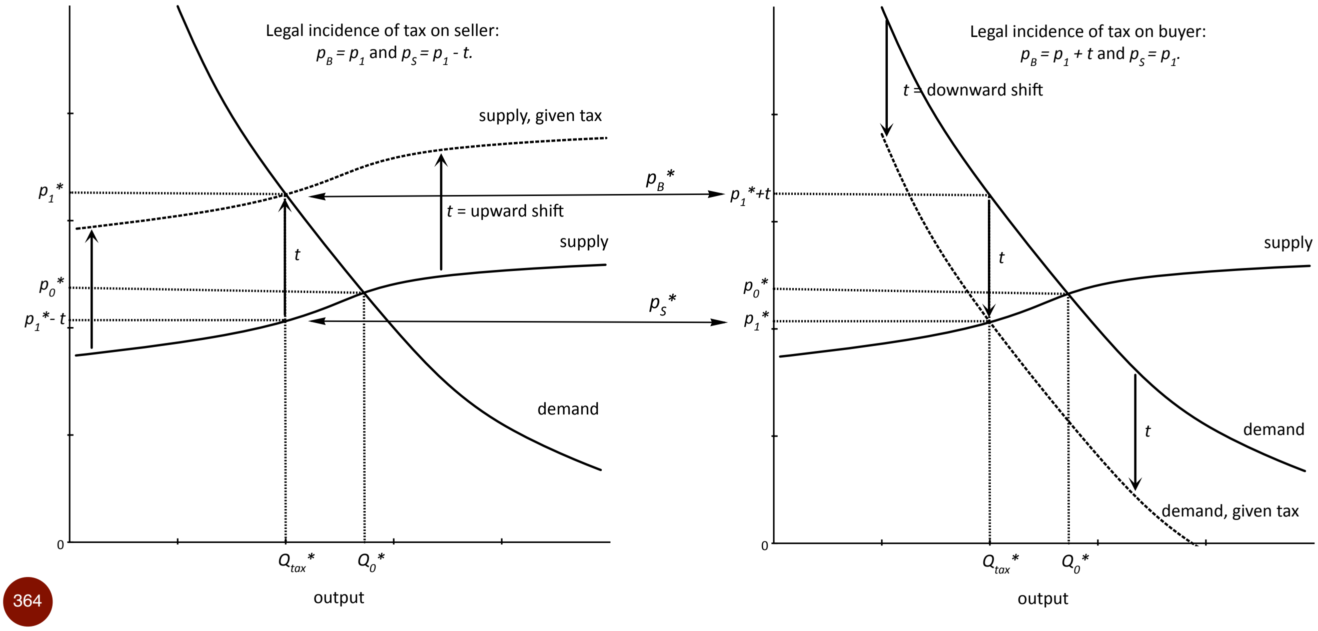
The new equilibrium price is p_1^* which here is the amount the seller receives, $p_1^* = p_s^*$. The buyer pays (including tax), $p_b^* = p_1^* + t$.

In this example, the buyer has the larger economic incidence of the tax.

Indeed, the economic incidence looks exactly the same as when the tax was imposed on the sellers.

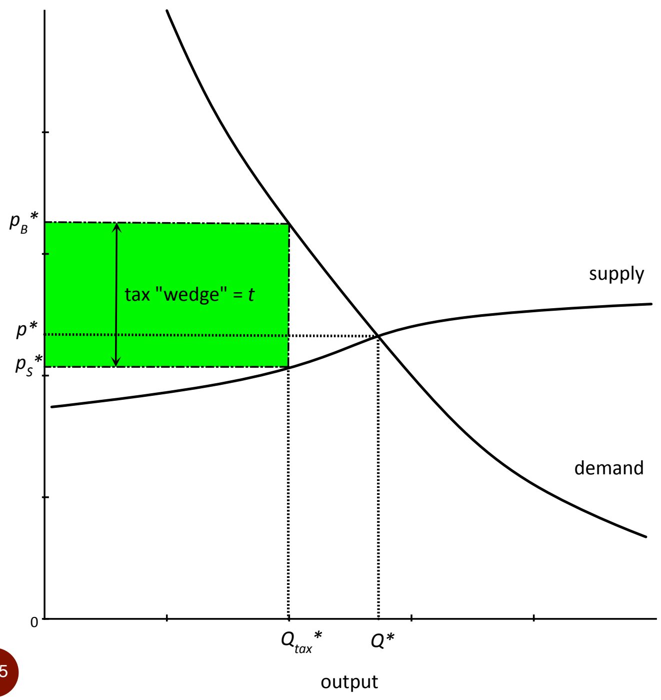
Geometry of econ incidence, continued

- Here we can clearly see that the tax-inclusive equilibrium prices are the same whether one legally imposes the tax on the sellers or the buyers.



Thinking about a unit tax as a wedge

- Given a supply and demand graph and the amount of unit tax, t, we can determine the after-tax equilibrium prices without shifts.



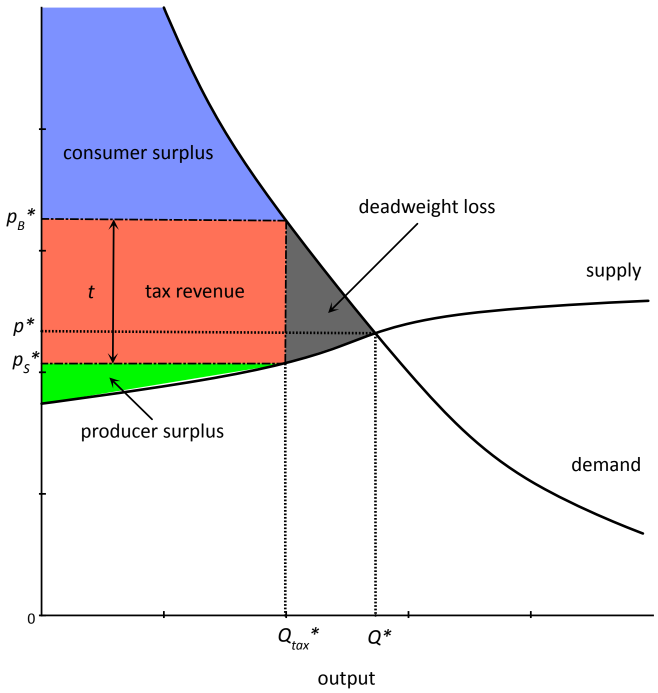
The tax-revenue rectangle in the graph is *t* dollars high.

If a rectangle of height *t* is horizontally inserted into a supply and demand graph such that the demand curve touches the top corner and the supply curve touches the bottom corner, then we are guaranteed the prices at the top and bottom edges, p_b* and p_s*, satisfy

$$p_b^* - p_s^* = t \text{ (rectangle height)},$$
and
$$D(p_b^*) = S(p_s^*) = Q_{tax}^*.$$

Deadweight loss (or excess burden) of tax

- To avoid shifting curves, let's just use a tax wedge to think about changes in surplus.



Consumer surplus is now defined with respect to the buyer's after-tax price; similarly, producer surplus is defined relative to the seller's after-tax price.

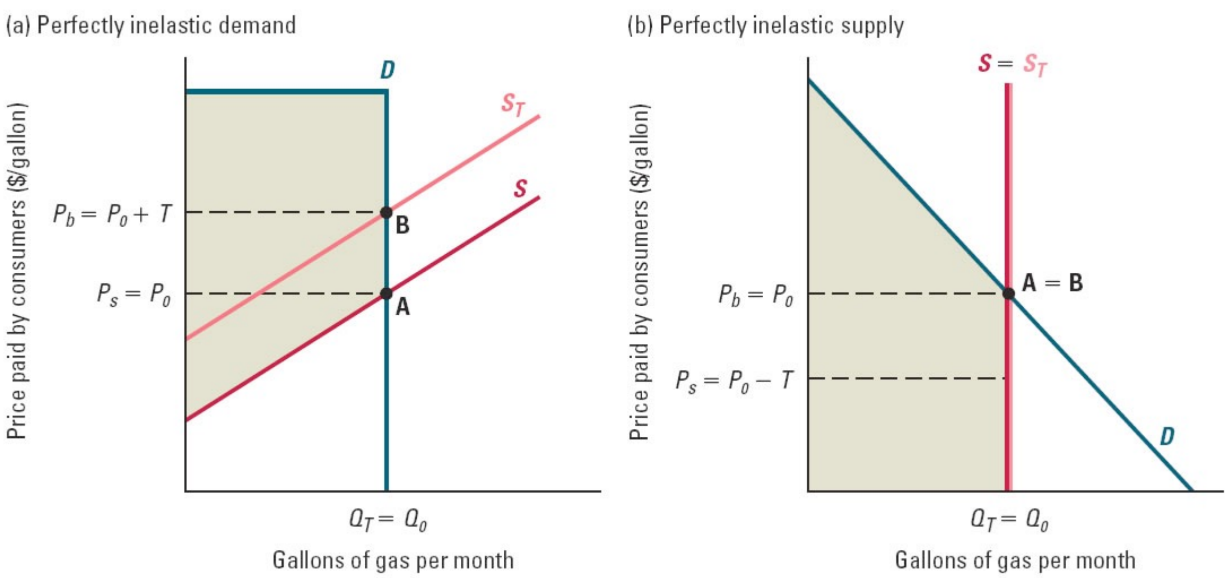
The deadweight loss is in the region between Q_0* and Q_{tax}*.

It represents all previous trades for which the joint surplus was positive but less than the tax, *t*. These are efficient trades that do not generate enough surplus to pay the tax, and so do not take place once the tax is imposed.

Less distortionary taxes?

- If we must use some taxes to pay for government and we want to raise the necessary tax revenue with as small a deadweight loss as possible, what kinds of goods should be taxed?
- Clearly, if a good is either perfectly inelastic on the demand or on the supply side, there is no deadweight loss. This is because Q_0*=Q_{tax}*, and no trades are discouraged.
  - Of course, in such a setting the inelastic side of the market bears the entire economic burden of the tax.
- More generally, economists desire to tax goods for which there will not be a large change in trade.
- Sometimes discouraging economic activity is efficient (when there are "negative externalities"). In this case, we can raise tax revenues while also increasing economic efficiency. We will discuss this idea in more detail below.

Taxation with No Deadweight Loss



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

How do slopes relate to incidence?

- Let's suppose that demand and supply are linear functions, but possibly with very different absolute slopes. (For small tax changes, assuming linear curves is immaterial.)
- Define as before

$$\Delta p_b = p_b^* - p_0^*$$

$$\Delta p_s = p_0^* - p_s^*.$$

Define the reduction in equilibrium output caused by the tax as

$$\Delta Q^* = Q_0^* - Q_{tax}^* \geq 0.$$

- As the figures make clear, linear curves imply

$$\Delta p_b = \Delta Q^* \cdot (\text{- slope of demand curve})$$

$$\Delta p_s = \Delta Q^* \cdot (\text{slope of supply curve}).$$

How do slopes relate to incidence?

- Many conclusions follow from

  $\Delta p_b = \Delta Q^* \cdot$ (- slope of demand curve)

  $\Delta p_s = \Delta Q^* \cdot$ (slope of supply curve).

- First, the relative size of the economic incidences is

$$\frac{\Delta p_b}{\Delta p_s} = \frac{|\text{slope of demand}|}{\text{slope of supply}} = \frac{1/|\varepsilon^d|}{1/\varepsilon_s} = \frac{\varepsilon^s}{|\varepsilon_d|}$$

- In fractional terms,

$$\frac{\Delta p_b}{\Delta p_b + \Delta p_s} = \frac{|\text{slope of demand}|}{|\text{slope of demand}| + \text{slope of supply}} = \frac{\varepsilon^s}{|\varepsilon_d| + \varepsilon^s}$$

- The reduction in quantity from the tax is

$$\Delta Q = \frac{t}{|\text{slope of demand}| + \text{slope of supply}} = \frac{t \cdot Q}{p}\left(\frac{|\varepsilon_d|\varepsilon^s}{|\varepsilon_d| + \varepsilon^s}\right)$$

Less distortionary taxes?

- A common mistake made by governments when introducing new taxes is to incorrectly think the market being taxed is inelastic.
  - Example. In the early 1990's luxury boats were taxed. Luxury boats (broadly defined) are sold to customers with inelastic demand. Unfortunately, the tax applied only to new boats purchased in the US. A middleman could buy a yacht from Canada and resell it to a US citizen and avoid the tax. Given the tax dollars at stake, "used foreign luxury boats" became a close substitute to "taxable new US luxury boats". New US luxury boats have very elastic demand. The tax was quickly repealed once the US boating industry went into a tailspin.
  - Example. State governments often raise their cigarette tax and learn that the demand for cigarettes is much more elastic than they anticipated. In reality, there are often two kinds of cigarettes -- those with high in-state taxes and cigarettes from neighboring states that have a lower tax.

Value-added (ad valorem) taxes

- What changes in the analysis if we change from unit/specific taxes to proportional/VAT taxes?
- As before, to raise a fixed amount of tax revenue, the after-tax prices are independent of whether the tax is legally imposed on buyers or sellers. Economic incidences and the equilibrium quantity traded depends only on supply and demand.
- Similarly,
  - if one side of the market is perfectly inelastic, that side bears the entire tax burden;
  - if one side of the market is perfectly elastic, that side bears none of the tax burden.
- For less extreme cases, the formulas for economic incidence are more complicated than with the unit tax, but the economic intuitions are mostly the same.

General equilibrium analysis of taxes

- General equilibrium analysis provides more accurate answers than partial equilibrium analysis, but at the cost of complexity.
- Consider the effects of a unit tax on ice cream
  - Assume pie and ice cream are complements
  - Assume no supply linkages
- Partial equilibrium considers the impact of a tax on reducing the consumption of ice cream, holding the price of pie fixed.
- General equilibrium analysis determines the equilibrium prices and quantities for ice cream and pie simultaneously.

General equilibrium analysis of taxes

- General equilibrium tax analysis also includes the indirect effect from the change in the price of pie:
  - when the tax increases the price of ice cream, the demand for pie shifts inward and the price of pie falls.
  - A reduction in the price of pie, however, causes the demand for ice cream to shift outward.
  - The outward shift in the demand for ice cream implies an even higher post-tax price of ice cream. In this example, partial equilibrium analysis understates the effect of a tax on the price of ice cream and overstates the reduction in ice cream consumed.

Subsidies (the reverse of taxes)

- Think of a subsidy, s, as a negative tax. The after-subsidy prices are still p_b and p_s, but now

$$p_s - p_b = s > 0$$

and each seller receives more per unit than the buyer pays.

- Using the same notation as before (but reversing signs)
  - the economic incidence for the buyer is

$$\Delta p_b = p_0^* - p_b^* \geq 0$$

  - the economic incidence for the seller is

$$\Delta p_s = p_s^* - p_0^* \geq 0$$

  - the total economic incidence equals the per-unit subsidy

$$\Delta p_b + \Delta p_s = s$$

  - and the change in output is $\Delta Q^* = Q_{sub}^* - Q_0^* \geq 0$.

Subsidies

- Given supply and demand functions which depend only on after-subsidy prices, market equilibrium requires

$$D(p_b)=S(p_s).$$

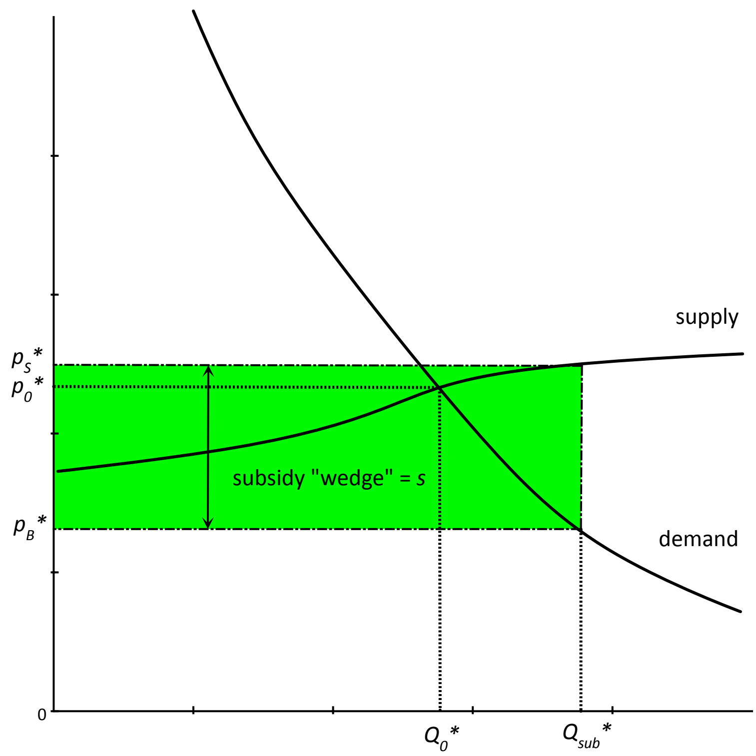
- Given the per unit subsidy is s, by definition we have

$$p_s - p_b = s.$$

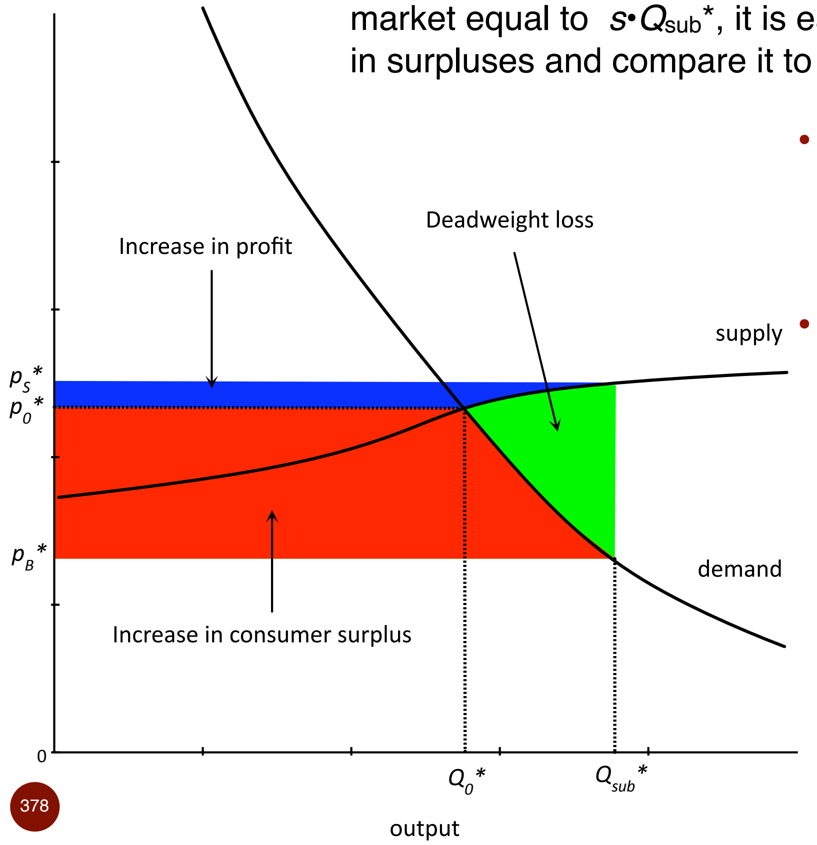
- Thus, the after-subsidy prices are entirely determined by the shape of supply and demand.
- Graphically, we can introduce a wedge of height *s* where the top edge gives us p_s and the bottom edge gives us p_b. The top edge is set to touch the supply curve (remember, it is p_s now) and the bottom edge is set to touch the demand curve. In such a configuration, the after-subsidy prices satisfy the two conditions above.

Subsidies

- More than the socially efficient level of output is produced.
- Benefits of subsidies go more to the inelastic side of the market.



Subsidies: surplus and deadweight loss



- Because we are injecting government revenues into the market equal to $s \cdot Q_{sub}$*, it is easier to denote the changes in surpluses and compare it to the subsidy cost.
- Note that the subsidy rectangle is larger than the increases in profit and consumer surplus by the amount of the *DWL*.
- Deadweight loss:
  - it is between Q_0* and Q_{sub}*
  - it represents trades that create a net social loss that is less than *s*, and hence will take place because the subsidy covers the loss.

Subsidies: formulas, etc.

- Given our use of notation, we continue to have

$$\Delta p_b = \Delta Q^* \cdot (\text{- slope of demand curve})$$

$$\Delta p_s = \Delta Q^* \cdot (\text{slope of supply curve}).$$

Therefore, the formula are the same as in the case of unit taxes:

- First, the relative size of the economic incidences is

$$\frac{\Delta p_b}{\Delta p_s} = \frac{|\text{slope of demand}|}{\text{slope of supply}} = \frac{1/|\varepsilon^d|}{1/\varepsilon_s} = \frac{\varepsilon^s}{|\varepsilon_d|}$$

- In fractional terms,

$$\frac{\Delta p_b}{\Delta p_b + \Delta p_s} = \frac{|\text{slope of demand}|}{|\text{slope of demand}| + \text{slope of supply}} = \frac{\varepsilon^s}{|\varepsilon_d| + \varepsilon^s}$$

- The increase in quantity from the subsidy is

$$\Delta Q = \frac{s}{|\text{slope of demand}| + \text{slope of supply}} = \frac{s \cdot Q}{p} \left(\frac{|\varepsilon^d| \cdot \varepsilon^s}{|\varepsilon^d| + \varepsilon^s} \right)$$

Day 7:
Applications in competitive markets - externalities, price controls, import quotas; basic monopoly

Externalities

- An externality is a benefit or cost that is not included in the decision making of the parties that generate it.
- In a competitive market, this implies that the benefit or cost is external to the market price.
  - A negative externality is an external cost that is imposed on people who are not part of the market transaction which generates the cost.
    - E.g., a coal-fired power plant in Illinois generates pollution that negatively impacts people who are not buyers of the electricity. If the plant does not pay for the pollution harm it causes, then its pollution is a negative externality.
  - A positive externality is an external benefit that is conferred on people who are not part of the market transaction which generates it.
    - E.g., when my neighbor buys flowers for his garden, I am not involved in any market transaction, but I still get to enjoy the flowers.

Examples of externalities

- Negative externalities:
  - pollution (air, noise, carbon, sight, etc.)
  - overuse of common resources (e.g., traffic congestion on streets, overfishing of oceans, tourists, etc.)
  - clearing a rain forest for cattle ranching
  - unhealthy behavior when health costs are paid in part by society (e.g., smoking, drinking, obesity)
  - dangerous behavior that affects others (e.g., driving fast or while intoxicated)
- Positive externalities:
  - generating and sharing ideas at below cost (blogging, privately-funded University research, etc.)
  - improving your looks (your self, home, garden, etc.)
  - education (e.g., better citizens, voters, etc.)
  - private, shared resources (private museums & libraries)

Externalities

- Competitive markets are generally efficient because the market demand curve reflects the marginal valuations of all consumers in society and the supply curve reflects the marginal costs of the producers in society. Hence, the market equilibrium maximizes society's benefits minus costs.
- With externalities, the market demand curve and the supply curve no longer reflect all social values and costs.
  - Unresolved market externalities lead to inefficient consumption and production.
- Two solutions to the problem of externalities:
  - [Pigou] Use taxes or subsidies to include the external effects (e.g., tax pollution, subsidy planting flowers, etc.)
  - [Coase] Define property rights and form markets for the external effects.

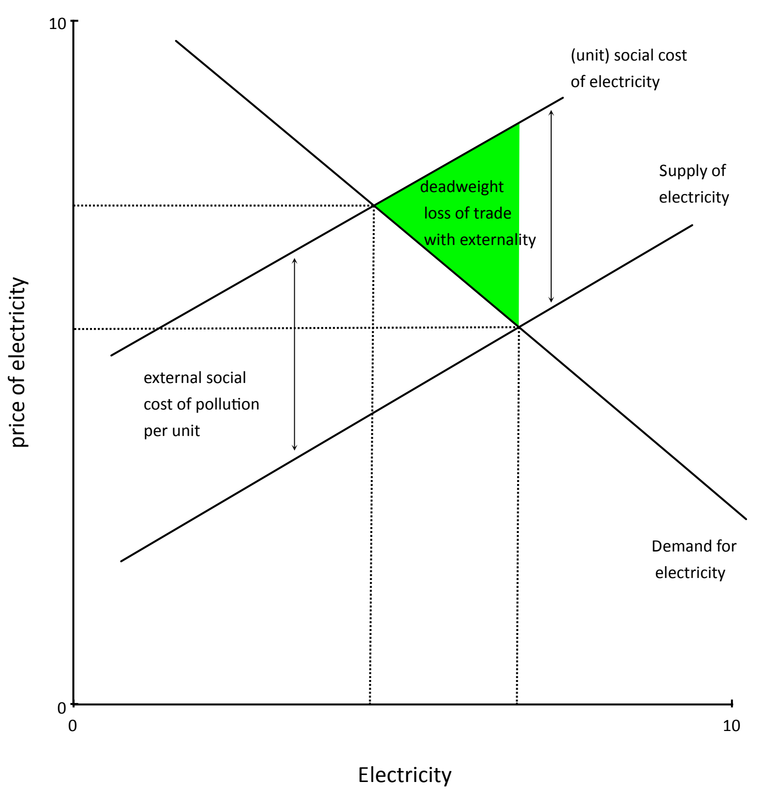
  In both cases, the externalities become internalized.

Pigou's approach to externalities

- Consider the case of a negative externality (e.g., pollution).
- The market equilibrium without any intervention is such that the last trade generates slightly more value to the buyer than it costs the producer to make. But suppose that the unit also imposes a negative cost of $1 on outsiders to the trade. Then the marginal trade actually destroys almost $1 of social value. We want to discourage such socially inefficient trades.
- If we make the buyer and seller collectively pay $1 in tax to the government, they will only trade if they generate at least $1 in surplus between themselves. But this will guarantee that only efficient trades take place.
- The Pigouvian tax (after Professor A.C. Pigou) is set to exactly equal the social cost of the trade. With such a tax imposed, the resulting market supply and demand curves will correctly internalize the external effects of trade.
  - An aside: One historian has argued that Pigou (a Professor in Cambridge, England) was also a spy for the Soviet Union during WWII and the years that followed (one of the "Cambridge Five.")

Pigou's approach to externalities

- Consider case of pollution externality:



Without internalizing the cost of pollution, there is a deadweight loss from too much trade.

A tax exactly equal to the external social cost of pollution causes the new equilibrium output to move to the efficient point.

Pigou's approach to externalities

- There are a few practical problems with solving externalities with taxes or subsidies:
  - How do we correctly determine the social cost or benefit of some activity? You would like each member of society to tell you how much they would individually pay to reduce pollution. But each person has an incentive to overstate this amount if asked.
  - Even if the government can correctly determine these amounts, why do we think the government will actually implement the correct taxes or subsidies? Realize that special interest groups may have tremendous incentives to influence governments to use different "estimates" of the social costs or benefits.
- Is there another way to internalize an externality?

Coase's approach to externalities

- Coase's insight is that if you assign property rights over the externality and all people impacted by the activity can get together and efficiently negotiate, then the efficient outcome will arise. This is the "Coase Theorem."
- Consider the case of pollution, but suppose there is only a single individual who is negatively effected.
  - If you assign the right to pollute to the power plant, the individual can still come forward and offer the plant money to reduce production to efficient levels. E.g., if the plant was producing 1000 megawatts and each megawatt costs the person $1 in pollution, he could offer the firm $1 for each unit of output reduction below 1000 and the efficient production level would arise. There are also efficient offers that more evenly split the gains from pollution reduction between the firm and the individual.
  - If you give the individual the right to be free of pollution, then the firm can always offer the individual $1 per unit to allow him to pollute. Again, efficient trade emerges.

Coase's approach to externalities

- If property rights over externalities are well defined, then the only reason inefficient activity can take place is if bargaining fails to reach efficient outcomes. (E.g., there are bargaining costs, etc.)
- Bargaining costs are likely to be low if a competitive market of price takers can be formed to trade in the externality.
  - This is the idea behind markets in carbon permits. Of course, the total level of trading permits is established by the government, and again we have similar problems as with Pigouvian taxes.
  - In many markets, it is unlikely externality trading will work well because there are too many people impacted by the externalities and there are difficulties with hold-out and free-riding.
    - E.g., imagine a firm trying to buy the pollution permissions from thousands of people. If it must obtain 100 percent of their rights in order to pollute, it is unlikely to succeed, even when pollution is efficient.

Coase's approach to externalities

- Even when there are just a few people involved in the costs and benefits of the externality, when information about values and costs is private, efficient trading is unlikely.
  - Roger Myerson (U of Chicago, Dept of Econ) won his Nobel prize, in part, for proving that bargaining is generally inefficient when there is private information. (The other reason for his Nobel prize is his work on auction design.)

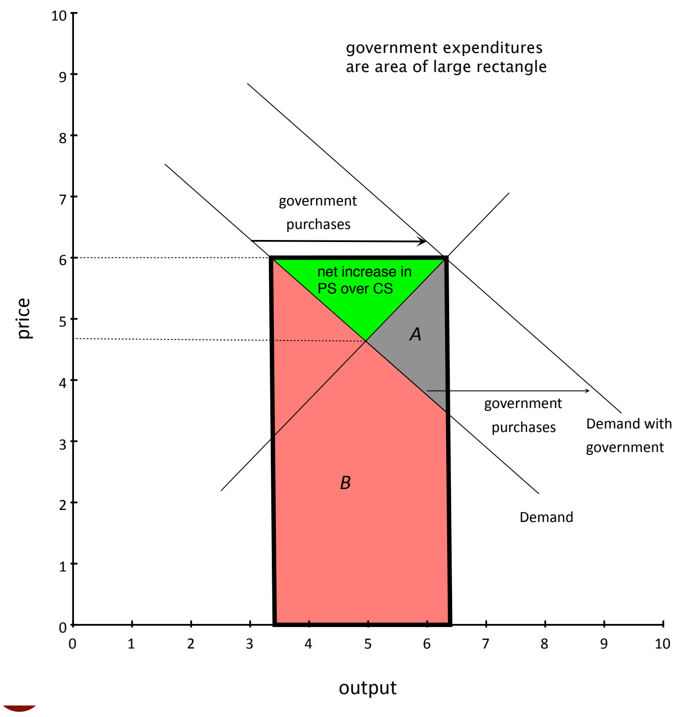
- At the end of the day, there is no perfect remedy for markets that suffer from externalities. The social costs of the externalities must be compared to the social costs of the imperfect remedies.

Topic 3: Other market interventions

- Price supports
- Price controls (e.g., minimum wage, rent control, etc.)
- Quantity controls (e.g., licenses to produce, quotas, etc.)
- Discriminatory controls
  - tariffs (e.g. taxes) on imports but not domestic production
    - subsidies on domestic production, but not imports, similar
  - quotas on imports, but not domestic production

Price supports

- A price support policy is a market intervention in which the government agrees to buy up output as needed in order to support a certain price.



To raise the price to $p=6$, the government buys up output.

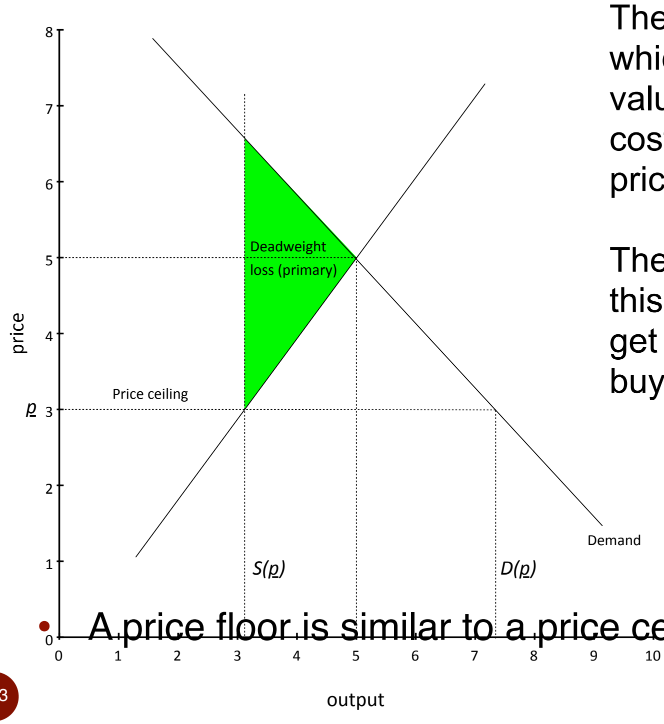
Too much is produced. If the government distributes the purchased output to the highest valued buyers who did not purchase, then the deadweight loss is area A. On the other extreme, if the purchased output is wasted, $DWL=A+B$.

Price controls

- A price control is a government mandate that all trade takes place at a legally fixed price.
- Unlike taxes, subsidies, and many other interventions, price controls break the market equilibrium: no longer does supply equal demand.
- When price controls are ceilings (e.g., rent controls, fuel price controls, etc.) that are lower than the uncontrolled equilibrium price, then we have $D(p) > S(p)$, and so only $S(p)$ units are traded and there is excess demand for output.
- When price controls are floors (e.g., minimum wage laws, etc.) that are higher than the uncontrolled equilibrium price, then we have $D(p) < S(p)$, and so only $D(p)$ units are traded and there is excess supply of output.

Price controls

- Consider a price ceiling and the resulting surpluses and DWL:



The primary DWL represents trades for which there exists a buyer with marginal valuation greater than a seller's marginal cost, but which cannot be supported by a price below the ceiling, p.

There is an additional deadweight loss to this amount if the buyers who are able to get the good are not the highest valuation buyers. This is not depicted in the graph.

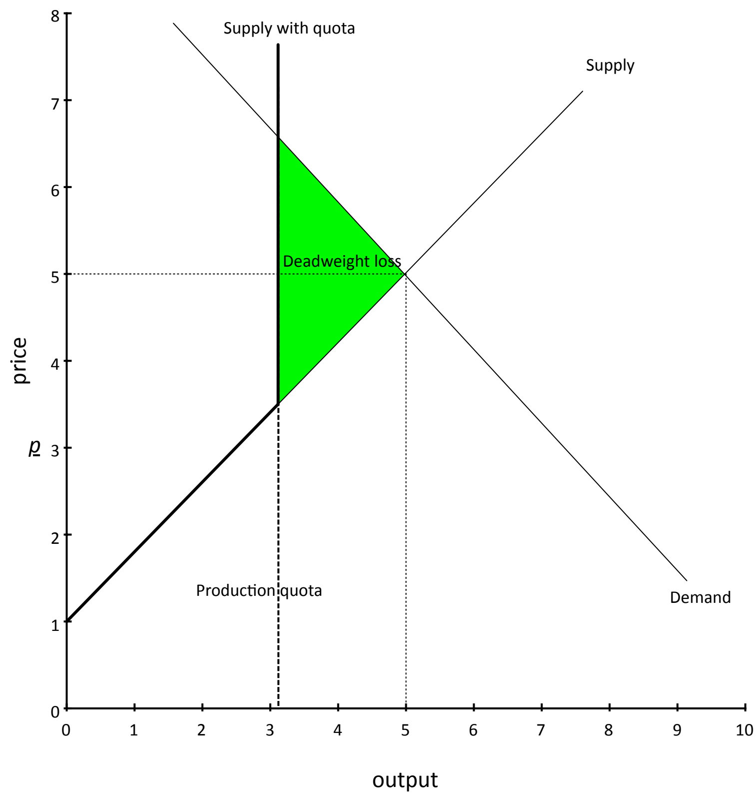
- A price floor is similar to a price ceiling, with a reverse graph.

Quantity controls

- We have already seen quantity controls in the context of the taxicab licensing example.
- Restrictions on output serve to raise prices and profits for those who are allowed to produce.
- Consumer surplus is lower and deadweight loss exists because there are some buyers who do not consume the good even though they value it more than the marginal cost of production.

Quantity controls

- Example of a production quota with upward-sloping supply:



Discriminating controls

- Until now, we have considered price and quantity controls that impacted the market uniformly.
- We now consider market controls that only apply to a segment of buyers or sellers. The most common division is with respect to domestic and foreign trade.
  - Import quotas are quotas that only apply to imported goods.
  - Tariffs are taxes which only apply to imported goods.

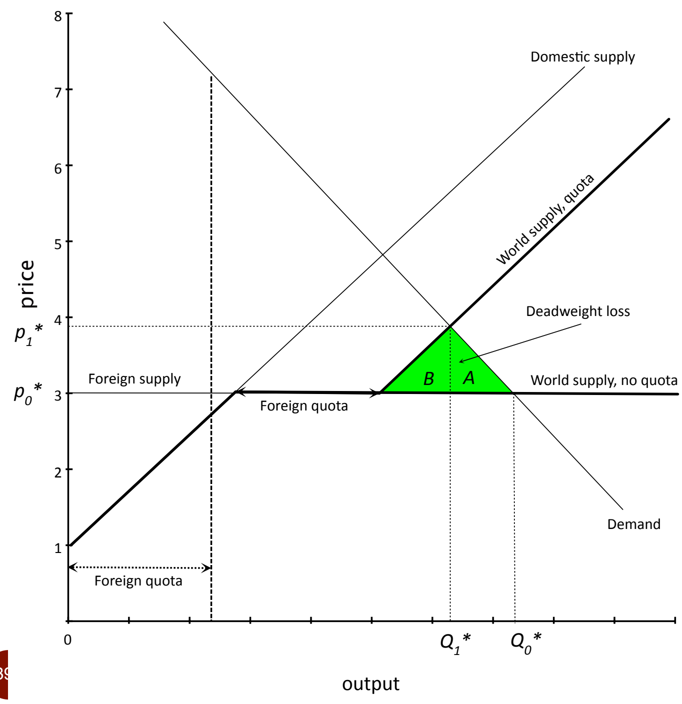
  In this line of inquiry, we could also consider domestic subsidies, import price floors, etc., but we will focus our attention on import quotas and tariffs.

Import quotas

- We assume that both foreign and domestic producers are price takers.
- The effect of a quota restriction on imports, Q_F, is to truncate the foreign supply curve at Q_F, making it vertical.
- Impact of import quotas:
  - profits increase for domestic producers
  - profits increase for foreign producers who continue to export under the quota
  - consumer surplus decreases for domestic consumers
  - deadweight loss from two sources:
    - too little is produced (some consumers do not buy even though their value exceeds the marginal cost of foreign producers)
    - wrong firms produce (some output is produced by domestic firms at a higher marginal cost than non-exporting foreign firms).

Import quotas

- For simplicity, suppose that foreign production is a constant-cost industry, but that domestic supply is upward sloping.



The deadweight loss from some consumers not buying even though values exceed marginal cost of foreign firms is the area in *A*.

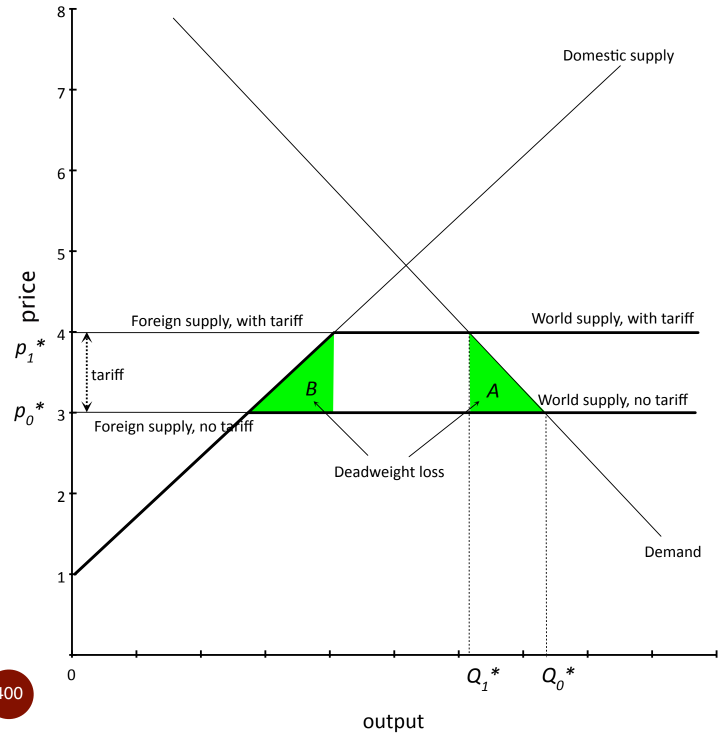
The deadweight loss from high-cost firms producing rather than low cost firms is the area in *B*.

Import tariffs

- We assume that foreign and domestic producers are price takers.
- The effect of a specific/unit tariff on imports, *t* , is to shift the foreign supply curve upwards by exactly the amount of the tariff.
- Impact of import tariffs:
  - profits increase for domestic producers
  - profits decrease for foreign producers
  - government earns tax revenue
  - consumer surplus decreases
  - deadweight loss from two sources:
    - too little is produced (some consumers do not buy even though their value exceeds the marginal cost of foreign producers)
    - wrong firms produce (some output is produced by domestic firms at a higher marginal cost than foreign firms).

Import tariffs

- For simplicity, suppose that foreign production is a constant-cost industry, but that domestic supply is upward sloping.



The deadweight loss from some consumers not buying even though values exceed marginal cost of foreign firms is the area in *A*.

The deadweight loss from high-cost firms producing rather than low cost firms is the area in *B*.

Basic model of monopoly pricing

- When a firm has market power, we mean that the demand curve for the firm's product is downward-sloping. The firm can choose its price within a range and is NOT a price taker.
  - In a competitive model, the demand curve for each firm's product is perfectly elastic even though the market demand curve is not. (E.g., agricultural commodities)
- In the standard monopoly model
  - there is a single firm facing a downward-sloping market demand curve for its product
  - there is no strategic interaction with firms producing related products.

Basic model of monopoly pricing

- How do monopolies arise?
  - Patents on technology or control of other essential inputs
  - Government regulation restricting entry
  - Market size can only support the fixed costs of one firm
  - Significant economies of scale (i.e., natural monopoly)

Basic model of monopoly pricing

- We assume for now that the monopolist must sell all output at the same price.
  - we will relax this assumption of one price for all when we discuss price discrimination later
- A monopolist will choose a point on the market demand curve to maximize profits:
  - it is wasteful to choose a price and output such that not all output is sold (should produce less)
  - it is wasteful to choose a price and output such that all of demand cannot be satisfied (should raise price).

Basic model of monopoly pricing

- Given the monopolist will always choose an output-price pair that is on the market demand curve, we can write revenue as a function of output by substituting in the demand curve that requires $p = P(q)$:

$$R(q) = p\,q = q\,P(q)$$

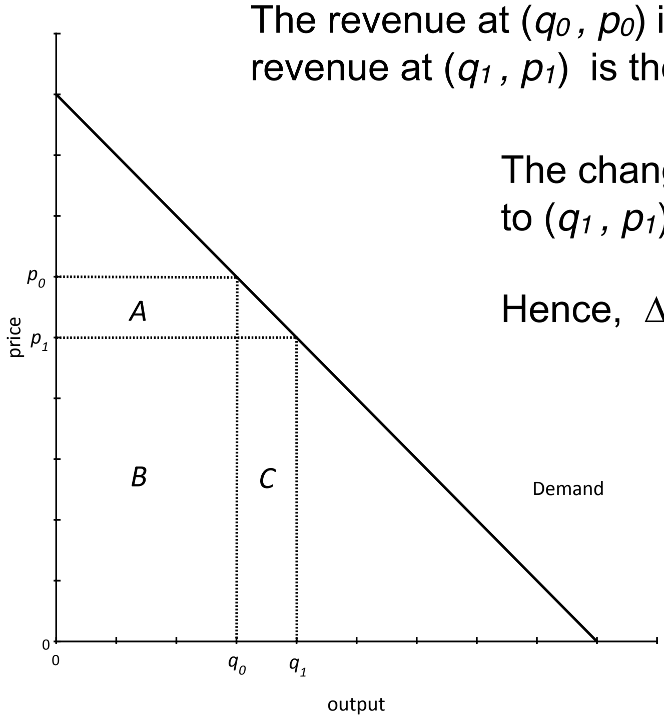
  - Note: you can also write revenue as a function of price by substituting $q = D(p)$ for q. We will not do this, however.

- In order to apply the marginal output rule, we will need to calculate marginal revenue. Let's do this first graphically, then using calculus. (This analysis should be familiar to you from Chicago when we discussed market revenues.)

Basic model of monopoly pricing

- Consider a change from an output sold of q_0 at price $p_0 = P(q_0)$ to a higher output of q_1 sold at a lower price of $p_1 = P(q_1)$.

The revenue at (q_0, p_0) is the area of the regions $A+B$; revenue at (q_1, p_1) is the area of the regions $B+C$.



The change in revenue going from (q_0, p_0) to (q_1, p_1) is $(B+C) - (A+B) = C-A$.

Hence, $\Delta R = p_1\,\Delta q + q_0\,\Delta p$, where

$$\Delta R = p_1 q_1 - p_0 q_0$$
$$\Delta q = q_1 - q_0$$
$$\Delta p = p_1 - p_0$$

Basic model of monopoly pricing

- The change in revenue in the figure is given by

$$\Delta R = p_1 \Delta q + q_0 \Delta p.$$

- The marginal revenue is the change in revenues per change in output, therefore we divide by Δq to obtain

$$MR = \Delta R/\Delta q = p_1 + q_0\, \Delta p/\Delta q.$$

- Selling an additional amount of output Δq at price p_1 directly contributes to revenue by p_1 per unit. Because price had to be reduced by Δp in order to sell those units, revenue is reduced on the original q_0 units by $q_0 \Delta p$ in total; this, revenue is reduced by $q_0 \Delta p/\Delta q$ per unit in the second term.
  - Note if demand is downward sloping and $q_0 > 0$, the second term is always negative, implying $MR < p_1$.

Basic model of monopoly pricing

- Now, let's redo the analysis using calculus. Revenue is given by

$$R(q) = p\, q = q\, P(q).$$

- Using calculus and the chain rule, marginal revenue is

$$MR(q) = P(q) + q\, P'(q).$$

- At positive output, the second term is negative (demand slopes downward), so MR is less than the price: $p > MR(q)$.
- The first term is the additional revenue the marginal unit brings in, $p=P(q)$; the second term is the revenue lost by lowering price on each of the q units by $P'(q)$ in order to sell the marginal unit.
  - The marginal output rule implies $MR(q) = MC(q)$, so

$$p > MR(q) = MC(q).$$

  Therefore, the market price exceeds marginal cost.

Basic model of monopoly pricing

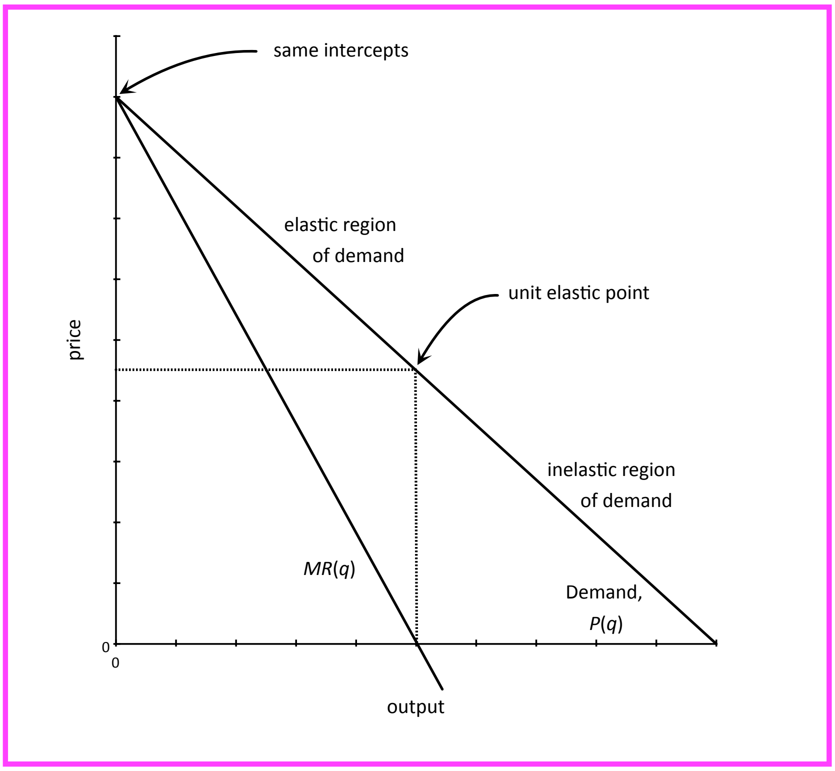
- Because marginal revenue is
$$MR(q) = P(q) + qP'(q)$$
we can write it in terms of the own-price demand elasticity
$$MR(q) = P(q)\left(1 + \frac{qP'(q)}{P(q)}\right) = P(q)\left(1 + \frac{1}{\varepsilon^d(q)}\right)$$
Because the elasticity is negative, *MR*(*q*) < *P*(*q*) remains true.
- Also note that
$$\begin{aligned} MR(q) > 0 \quad &\iff \quad |\varepsilon^d(q)| > 1, \text{elastic demand} \\ MR(q) < 0 \quad &\iff \quad |\varepsilon^d(q)| < 1, \text{inelastic demand.} \end{aligned}$$
- Finally, note that perfectly elastic demand implies *MR*(*q*) =*p*. The competitive model with price-taking is a special case!

Basic model of monopoly pricing

- Let's graph the marginal revenue function for the case of linear demand, using
$$P(q) = a - bq.$$
- Our general formula
$$MR(q) = P(q) + qP'(q)$$
becomes
$$MR(q) = (a - bq) + q(-b) = a - 2bq.$$
- Remarkably, with linear demand the marginal revenue curve has the same intercept on the price axis but twice the slope.
- In the general, nonlinear case, we have similar properties:
  - *MR*(*q*) has the same vertical intercept as *P*(*q*). Why?
    - Marginal revenue for the first unit sold is always the price of that unit. There can be no price reductions for other units.
  - q>0 implies P(q)>MR(q), so MR decreases faster, in general.

Basic model of monopoly pricing

- Graph of linear demand and *MR* curve. Note that both marginal revenue, $MR(q)$, and price, $P(q)$, are measured in money/unit, so both can be placed in the same graph.



Basic model of monopoly pricing

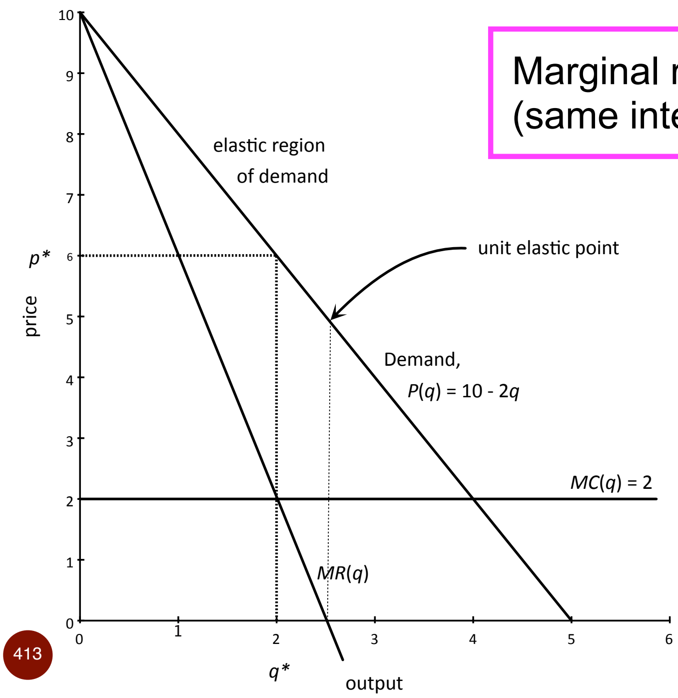
- We now take the monopolist's marginal revenue function and apply to it the two rules of profit maximization:
  - marginal output rule: if $q^* > 0$ is optimal, then
    $$MR(q^*) = MC(q^*);$$
  - shutdown rule: operate only if $\pi(q^*) \geq 0$, or equivalently,
    $$p^* = P(q^*) \geq AC^O(q^*).$$
- With monopoly, we generally skip over the shutdown rule because an industry which cannot support even one firm is not very interesting. That said, the rule must still be applied.

Basic model of monopoly pricing

- Optimal choice of (q^*,p^*).
  - Step 1: after constructing the monopolist's revenue as a function of output, we find the output which satisfies the marginal output rule:
    $$MR(q^*) = MC(q^*);$$
  - Step 2: Using q^*, we compute the corresponding optimal price using the demand curve.
    $$p^* = P(q^*);$$
  - Step 3 (typically skipped in class): we check that the firm should indeed operate at this (q^*,p^*). We confirm that
    $$p^* \geq AC(q^*),$$
    where $AC(q^*)$ is the average **opportunity** cost at q^*.
- If there are multiple decision horizons, we apply these 3 steps to each time frame, using the appropriate unit cost functions.

Basic model of monopoly pricing

- Example with linear demand and constant marginal costs.
  - Demand: $P(q) = 10 - 2q.$
  - Cost: $C(q) = 2q,$ therefore, $MC(q)$=2.



Marginal revenue is $MR(q) = 10 - 4q$
(same intercept as demand, but twice slope)

Marginal output rule:
$$MR(q) = MC(q)$$
$$10 - 4q = 2.$$
The optimal output is q^*=2; the corresponding optimal price is
$$p^* = P(q^*) = 10 - 2(2) = 6.$$

Because, $p^* = 6 \geq AC(q^*) = 2,$ shutdown is not optimal.

Basic model of monopoly pricing

- Let's focus our attention on the marginal output rule.

$$MR(q^*) = MC(q^*).$$

- In elasticity terms,

$$MR(q) = P(q)\left(1 + \frac{1}{\varepsilon^d(q)}\right)$$

so the marginal output rule can be written as

$$P(q^*)\left(1 + \frac{1}{\varepsilon^d(q^*)}\right) = MC(q^*)$$

- From here, we can deduce both

$$\frac{p^*}{MC(q^*)} = \frac{\varepsilon^d(q^*)}{1+\varepsilon^d(q^*)} \quad \text{and} \quad \frac{p^* - MC(q^*)}{p^*} = -\frac{1}{\varepsilon^d(q^*)}.$$

Basic model of monopoly pricing

- The first restatement of the marginal output rule

$$\frac{p^*}{MC(q^*)} = \frac{\varepsilon^d(q^*)}{1+\varepsilon^d(q^*)}$$

tells us that the ratio of price to marginal cost depends only on the elasticity of demand.

  - The more elastic the market, the lower the markup:
    - at ε^d = -2, the optimal ratio is 2 (i.e., price is twice MC)
    - at ε^d = -11, the optimal ratio is only 1.1 (price is only 10 percent more than marginal cost).
  - What happens if the demand is inelastic, $|\varepsilon^d| < 1$?
    - For example, if ε^d = -0.5, then the optimal ratio of price to marginal cost is negative. This cannot be correct!
- A monopolist always chooses a price in the elastic region of demand. Why?

Basic model of monopoly pricing

- A monopolist always chooses a price in the elastic region of demand. To understand why, suppose otherwise.
- At any point on the inelastic region of the demand curve, profits can be increased by reducing output and raising price.
  - By reducing output a small amount, the monopolist saves on production. Assuming that marginal cost is either zero or positive, this implies a reduction in total costs.
  - By reducing output a small amount, the monopolist also raises price accordingly and moves up the demand curve. If demand is inelastic, $MR(q) < 0$, so revenues increase!
  - If costs decrease and revenues increase, profits must increase. Hence, from any point on the demand curve that is inelastic, a movement up the curve increases profits.
- Aside: Antitrust policy, the US Supreme Court's "cellophane fallacy."

Basic model of monopoly pricing

- The second restatement of the marginal output rule is

$$\frac{p^* - MC(q^*)}{p^*} = -\frac{1}{\varepsilon^d(q^*)}.$$

  This identifies the optimal percentage markup between price and marginal cost with the inverse of the demand elasticity.
- This form of the marginal output rule is called the "inverse-elasticity pricing rule." The lefthand side is sometimes referred to as the "Lerner index" of market power.
- As the demand curve for a firm's product becomes more elastic, the markup decreases. In the extreme, with perfectly elastic demand, the monopolist will choose to price at marginal cost. This is consistent with our competitive model.

Basic model of monopoly pricing

- We want to illustrate consumer surplus, profit and deadweight loss in the case of monopoly pricing.
  - consumer surplus is the area under the demand curve and above the monopoly price
  - economic profit can be illustrated in a few different ways; the simplest way is to note that

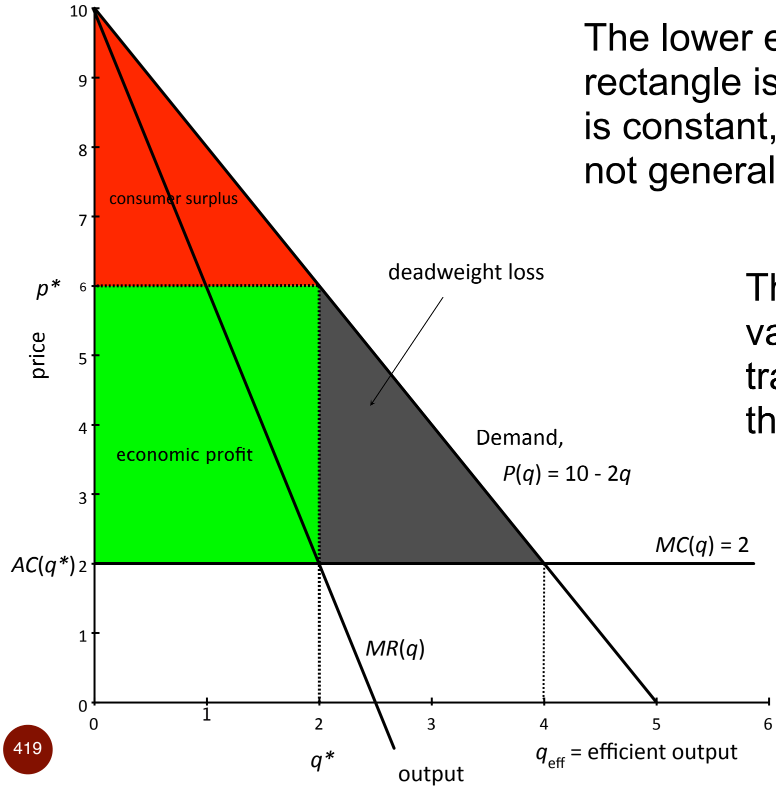
    $$\pi(q) = q(p - AC(q)),$$

    which is the area of a rectangle that is q units long and

    (p-$AC(q)$) dollars/unit high.
  - Deadweight loss is the area between the demand curve and the marginal cost curve, and between the monopoly output and the efficient output (which maximizes welfare).
    - Usual story: insufficient trade takes place.

Basic model of monopoly pricing

- Example of consumer surplus and profit with constant $AC(q)$.

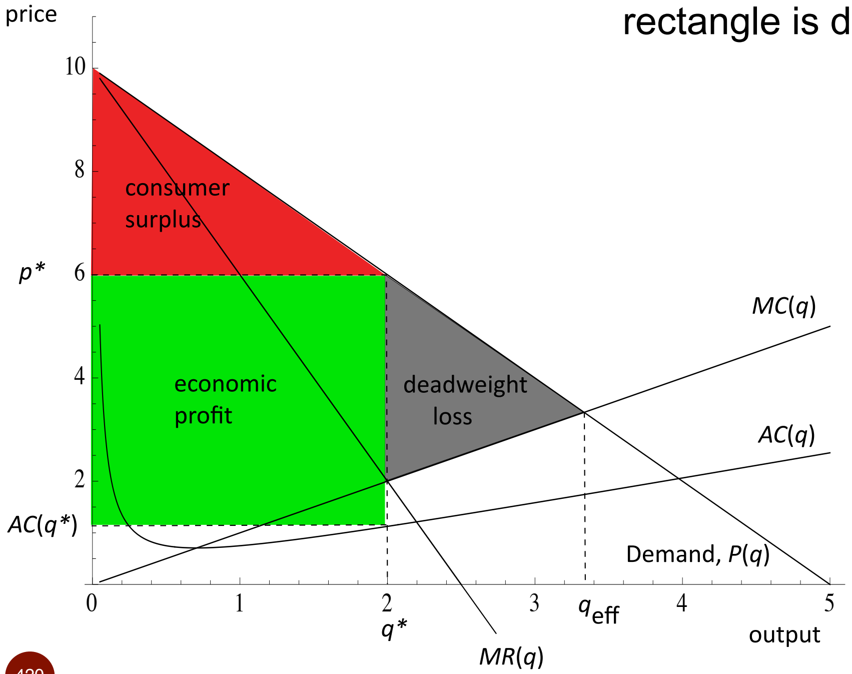


The lower edge of the economic profit rectangle is defined by $AC(q^*)$. If $AC(q)$ is constant, $AC(q^*)=MC(q^*)$, but this is not generally true.

The deadweight loss is the value of surplus-generating trades that do not take place at the monopoly price.

Basic model of monopoly pricing

- Example with non-constant $AC(q)$: $C(q) = 0.5q^2+0.25$.



The lower edge of the economic profit rectangle is defined by AC(q*).

The deadweight loss is the value of surplus-generating trades that do not take place at the monopoly price.

Note: The lower boundary of the DWL is the $MC(q)$ and not $AC(q)$.

Basic model of monopoly pricing

- Taxicab example: suppose now that one firm owns all 6 licenses.
  - Recall that $P(q) = 1 - q/(100{,}000)$, thus
    $$MR(q) = 1 - q/(50{,}000).$$
  - Marginal cost is 0.20 per mile if operating below capacity.
- Marginal output rule implies
  $$MR(q) = 1 - q/(50{,}000) = MC(q) = 0.20,$$
  so $q^* = 40{,}000$ miles at price $p^* = 1-(40{,}000)/(100{,}000) = 0.60$.
- Consumer surplus is
  $$CS = (1/2)(1 - 0.60)(40{,}000) = 8{,}000 \text{ dollars.}$$
- Profit (ignoring license cost) is $\pi = (40{,}000)(0.60-0.20) = 16{,}000$.
- DWL = (1/2)(80,000-40,000)(0.60-0.20) = 8,000 dollars.
  - Alternatively, DWL = maximum welfare - actual welfare
    $$\text{DWL} = 32{,}000 - (8{,}000+16{,}000) = 8{,}000.$$

Basic model of monopoly pricing

- Quick "estimate" of deadweight loss when (1) demand is linear and (2) marginal costs are constant, $C(q) = F + MC \cdot q$.
- Because MC is constant, $q^* = (1/2)q^{eff}$.
  - This holds because the MR curve for linear demand bisects any horizontal line between the vertical axis and the demand curve.
- Deadweight loss with linear demand and constant MC is thus

$$\begin{aligned} DWL &= (1/2)(p^* - MC)(q^{eff} - q^*) \\ &= (1/2)(p^* - MC)\ q^* = (1/2)(R(q^*) - C(q^*) + F) \\ &= (1/2)(\pi(q^*) + F). \end{aligned}$$

- Thus, DWL is (1/2) of profit before subtracting fixed cost.
- Similar calculations for consumer surplus yield

$$CS = (1/2)(\pi(q^*) + F) = DWL.$$

- Review the previous taxicab example to see that this is true.

Basic model of monopoly pricing

- A final comment on the deadweight loss of monopoly (due to Justice/Professor Posner at the U of Chicago Law School):
  - Our measure of DWL assumes that the firm acquired the right to be a monopolist at no cost.
  - In many instances, however, there is a contest between firms to become the monopolist. Resources used to compete for the monopoly position that have no social value are DWL.
  - Resources may also be used to maintain a monopoly position
  - A complete measure of DWL should include these additional socially-wasteful expenditures.
  - One can go so far to argue that if all firms are equally likely to win the monopoly contest, then each will compete to the point where the expected profit is equal to zero.
  - In this case, monopoly profit is actually DWL.

Day 8:
Applications of monopoly model; game theory; oligopoly

Basic Monopoly model (review)

- Taxicab example: suppose now that one firm owns all 6 licenses.
  - Recall that $P(q) = 1 - q/(100{,}000)$, thus
    $$MR(q) = 1 - q/(50{,}000).$$
  - Marginal cost is 0.20 per mile if operating below capacity.
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Applications of monopoly model

- Application 1: Optimal markups based on elasticities
- Application 2: Pricing with multiple products
- Application 3: Taxes, subsidies, cost pass through
- Application 4: Generalizing with dynamic demand
- Application 5: Vertical relations and double margins
- Application 6: Choice of quality
- Application 7: Monopoly/dominant firm facing a competitive fringe of price-taking firms

Application 1: Optimal markups using elasticities

- Let's focus our attention on the marginal output rule.

$$MR(q^*) = MC(q^*).$$

- In elasticity terms,

$$MR(q) = P(q)\left(1+\frac{1}{\varepsilon^d(q)}\right)$$

so the marginal output rule can be written as

$$P(q^*)\left(1+\frac{1}{\varepsilon^d(q^*)}\right) = MC(q^*)$$

- From here, we can deduce both

$$\frac{p^*}{MC(q^*)} = \frac{\varepsilon^d(q^*)}{1+\varepsilon^d(q^*)} \quad \text{and} \quad \frac{p^* - MC(q^*)}{p^*} = -\frac{1}{\varepsilon^d(q^*)}.$$

Application 1: Optimal markups using elasticities

- The first restatement of the marginal output rule

$$\frac{p^*}{MC(q^*)} = \frac{\varepsilon^d(q^*)}{1+\varepsilon^d(q^*)}$$

tells us that the ratio of price to marginal cost depends only on the elasticity of demand.

  - The more elastic the market, the lower the markup:
    - at ε^d = -2, the optimal ratio is 2 (i.e., price is twice MC)
    - at ε^d = -11, the optimal ratio is only 1.1 (price is only 10 percent more than marginal cost).
  - What happens if the demand is inelastic, $|\varepsilon^d| < 1$?
    - For example, if ε^d = -0.5, then the optimal ratio of price to marginal cost is negative. This cannot be correct!
- A monopolist always chooses a price in the elastic region of demand. Why?

Application 2: Multi-product pricing

- Consider a monopolist selling two products, *x* and *y,* which are demand substitutes for each other

$$\text{Good X:} \quad x = D_x(p_x, p_y), \quad p_x = P_x(x, y),$$
$$\text{Good Y:} \quad y = D_y(p_x, p_y), \quad p_y = P_y(x, y).$$

- Suppose that the total cost of producing is simply

$$C(x, y) = c_x x + c_y y$$

- Total revenue is

$$R(x, y) = xP_x(x, y) + yP_y(x, y)$$

- Marginal revenue of good *x* is therefore

$$MR_x(x, y) = P_x(x, y) + x\frac{\partial P_x(x,y)}{\partial x} + y\frac{\partial P_y(x,y)}{\partial x}.$$

- Notice that there is a third term in the standard MR expression which indicates the loss to revenue from *y* caused by increased sales of *x*:

$$MR_x(x,y) = P_x(x,y) + x\frac{\partial P_x(x,y)}{\partial x} + y\frac{\partial P_y(x,y)}{\partial x}.$$

  This has the effect of decreasing *MR* and therefore inducing a higher optimal price for *x*.

- It is more convenient to consider the profit maximization problem of choosing *prices* for the two goods to maximize

$$\pi(p_x,p_y) = (p_x - c_x)D_x(p_x,p_y) + (p_y - c_y)D_y(p_x,p_y).$$

- Differentiating with respect to prices yields 2 conditions:

$$D_x(p_x,p_y) + (p_x - c_x)\frac{\partial D_x(p_x,p_y)}{\partial p_x} + (p_y - c_y)\frac{\partial D_y(p_x,p_y)}{\partial p_x} = 0,$$

$$D_y(p_x,p_y) + (p_y - c_y)\frac{\partial D_y(p_x,p_y)}{\partial p_y} + (p_x - c_x)\frac{\partial D_x(p_x,p_y)}{\partial p_y} = 0.$$

$$D_x(p_x,p_y) + (p_x - c_x)\frac{\partial D_x(p_x,p_y)}{\partial p_x} + (p_y - c_y)\frac{\partial D_y(p_x,p_y)}{\partial p_x} = 0,$$

$$D_y(p_x,p_y) + (p_y - c_y)\frac{\partial D_y(p_x,p_y)}{\partial p_y} + (p_x - c_x)\frac{\partial D_x(p_x,p_y)}{\partial p_y} = 0.$$

Simplifying, the first equation, we obtain

$$x\left(1 + \frac{p_x - c_x}{p_x}\frac{p_x}{x}\frac{\partial D_x(p_x,p_y)}{\partial p_x}\right) + \frac{\partial D_y(p_x,p_y)}{\partial p_x}\frac{p_x}{y}\frac{p_y - c_y}{p_x}y = 0.$$

Using the definitions of various elasticities, we have more simply

$$x\left(1 + \frac{p_x - c_x}{p_x}\varepsilon_{x,p_x}\right) + \left(\varepsilon_{y,p_x}\frac{y(p_y - c_y)}{p_x}\right) = 0.$$

The first part of this expression is the same as in the one-product case. The second term represents the marginal profit generated on product *y* by an increase in the price of *x*. A similar relationship holds for the optimal price for good *y*. Thus, the presence of demand substitutes requires an increase in the prices of each good.

Application 3: Taxes, subsidies, etc.

- Competitive markets:
  - A change in unit costs acts (e.g., tax) acts as a wedge between the demand and supply curves;
  - the legal incidence of a tax is irrelevant for determining the economic incidence;
  - a fraction between 0 and 1 of a change in unit costs (e.g., a unit tax or subsidy) is passed on to buyers.
- Monopoly:
  - A change in unit costs acts (e.g., tax) acts as a wedge between the marginal revenue and marginal cost curves
    - Because competitive supply curves are essentially the industry's "marginal cost" curve, the important change here is that the wedge acts on the MR curve rather than demand.

Application 3: Taxes, subsidies, etc.

- Monopoly:
  - A change in unit costs (e.g., tax) acts as a wedge between the firm's marginal revenue and marginal cost curves
    - Note that competitive supply curves are essentially the industry's "marginal cost" curve, so the critical change under monopoly is that the wedge acts on the *MR* curve and not the demand curve.
  - The legal incidence is irrelevant for economic incidence.
  - Because the $MR(q)$ curve has a different shape than the monopolist's demand curve $P(q)$, the fraction of pass-through to buyers can exceed 1.
    - With linear demand and constant *MC*, the fraction of pass-through is always 1/2.
    - With constant elasticity of demand, the pass through is

$$\frac{|\varepsilon^d|}{|\varepsilon^d| - 1} > 1.$$

Application 4: Dynamic demand

- Suppose that demand function at date t is given by

$$q_t = D(p_t, q_{t-1}),$$

where q_{t-1} is the amount consumed last period.

- The short-run and long-run demand elasticities in period t are

$$\begin{aligned} \varepsilon^d_{SR} &= \frac{p_t}{q_t}\left(\frac{\partial D(p_t, q_{t-1})}{\partial p_t}\right) \\ \varepsilon^d_{LR} &= \frac{p_t}{q_t}\left(\frac{\partial D(p_t, q_{t-1})}{\partial p_t}\right) \Big/ \left(1 - \frac{\partial D(p_t, q_{t-1})}{\partial q_{t-1}}\right) \end{aligned}$$

- If consumption last period increases demand this period, then

$$|\varepsilon^d_{LR}| > |\varepsilon^d_{SR}|.$$

Application 4: Dynamic demand

- What is the correct inverse-elasticity pricing formula?
- The answer requires that we maximize the present value of profits given that consumption last period was q_{t-1}.
- The general solution is quite messy and requires some tools of dynamic optimization.
- We can more easily determine what the price should be in steady state, where $q_t = q_{t-1} = q_\infty$ and the price is p_∞ .
- Let r be the interest rate for discounting profits. A dynamic version of the marginal output rule gives us

$$\frac{p_\infty - C'(q_\infty)}{p_\infty} = \frac{1}{|\bar{\varepsilon}^d|}, \quad \text{where } \bar{\varepsilon}^d = \left(\frac{r}{1+r}\right)\varepsilon^d_{SR} + \left(\frac{1}{1+r}\right)\varepsilon^d_{LR}.$$

- Note that the appropriate elasticity is a weighted average of the short- and long-run elasticities.

Application 5: Capturing surplus in vertical relationships

- Consider a simple setting in which an upstream monopoly producer sells to a downstream monopoly retailer.
  - Suppose that the demand curve for the retail product is

    $P(q) = 10 - 2q.$
  - The marginal and average costs of producing the wholesale product are equal to c_w.
  - The marginal and average costs of retail sales are equal to c_r.
- Suppose the producer sets a per-unit wholesale price, w, at which the retailer may purchase unlimited quantities.
- The retailer takes the wholesale price, w, as given and chooses the optimal retail price, p, and level of output, q. This is the same setting as a monopolist with marginal cost $w + c_r$.
- What happens?

Application 5: Vertical relations

- As a benchmark, let's first determine what would happen if the upstream and downstream firms merged to form a single vertically integrated firm.
- Profit for the vertically integrated firm (as a function of q) is

  $\pi_{vi} = (P(q) - c_w - c_r)q = (10 - 2q - c_w - c_r)q.$
- The optimal output choice satisfies MR(q)= MC(q), so

  $10 - 4q = c_w + c_r$
- The optimal output and price are

  $q_{vi}^* = (1/4)(10 - c_w - c_r),$ $p_{vi}^* = (1/2)(10 + c_w + c_r),$

  and the economic profit of the vertically-integrated firm is

  $\pi_{vi}^* = (1/8)(10 - c_w - c_r)^2.$

Application 5 : Vertical relations

- Suppose now the firms are not integrated.
- The monopoly retailer effectively has a marginal cost of $w + c_r$.
- Setting marginal revenue equal to marginal cost, the retailer solves

$$MR_r(q) = 10 - 4q = w + c_r = MC_r(q).$$

- Solving for the downstream firm's demand, we obtain a demand function which depends upon w:

$$q_r(w) = (1/4)(10 - (w+c_r)).$$

- Assuming that the wholesale monopolist anticipates this, the upstream firm chooses w to maximize profits:

$$\pi_w = q_r(w)(w - c_w) = (1/4)(10 - (w+c_r))(w - c_w).$$

- Maximizing with respect to the wholesale price, w, we have

$$w^* = (1/2)(10 + c_w - c_r).$$

Application 5: Vertical relations

- We conclude that the unintegrated wholesale price is

$$w^* = (1/2)(10 + c_w - c_r).$$

We can calculate the resulting output, retail price, and the economic profits of the retailer and wholesaler:

- Output and retail price:

$$q^* = (1/4)(10 - (w^*+c_r)) = (1/8)(10 - c_w - c_r)$$

which is half of the vertically-integrated output, and

$$p^* = (1/4)(30 + c_w + c_r),$$

which is substantially higher than the vertically-integrated price.

- Profits. Using w^*, q^* and p^*, the profits are

$$\pi_w^* = (1/16)(10 - c_w - c_r)^2 \text{ and } \pi_r^* = (1/32)(10 - c_w - c_r)^2$$

and so $\pi_w^* + \pi_r^* = (3/4)\,\pi_{vi}^*$.

Application 5: Vertical relations

- What we have shown is that the non-integrated structure has lower profits because prices are too high (from the perspective of maximizing joint profits).
- This is referred to as the problem of **double marginalization**.
- Both firms are introducing a deadweight loss to increase their own profits, but the deadweight loss of one firm negatively impacts the other firm's profits.
- **Related application:** Why do corrupt bureaucrats (i.e., those requesting bribes for approval) cause greater inefficiencies when they act independently than if all the bureaucrats could collectively set their bribes?

Application 5: Vertical relations

- Without vertical integration, two firms may be able to avoid the problem of double marginalization:
  - Resale price maintenance, RPM: wholesaler imposes a maximum price that the retailer can charge;
  - Two-part wholesale prices: wholesaler charges a franchise fee and a per-unit wholesale price $w = c_w$.

Application 6: Choice of Quality

- Suppose that the monopolist chooses not just p and q, but also a demand-related variable, x.
  - x might represent quality (like durability)
  - x might be expenditures on advertising that increases demand
  - x may be a design factor that is valued by some customers and disliked by others
- Demand can now be written as a demand function,

  $$q=D(p,x),$$

  or as an inverse-demand function,

  $$p=P(q,x).$$

Application 6: Choice of Quality

- Let's continue to solve the monopolist's problem by selecting q (rather than p) for convenience, but now the monopolist also selects x.
- Formally, the monopolist chooses (q,x) to maximize

  $$\pi(q,x) = P(q,x)q - C(q,x),$$

  where $C(q,x)$ is the opportunity cost of producing q units with factor x.
- The monopolist now has two marginal conditions:
  - marginal output rule: $P(q,x) + P_q(q,x)q = C_q(q,x)$,
  - marginal x rule: $P_x(q,x)q = C_x(q,x)$,

  where the subscripts represent partial derivatives.

Application 6: Choice of Quality

- Compare the monopolist's conditions
$$P(q,x) + P_q(q,x)q = C_q(q,x),$$
$$P_x(q,x)q = C_x(q,x).$$
with a social planner's ideal choices.
- Total surplus, as a function of (q,x) is
$$W(q,x) = \int_0^q P(z,x)dz - C(q,x).$$
- The marginal conditions that give the planner's ideal (q,x) are
$$P(q,x) = C_q(q,x),$$
$$\int_0^q P_x(z,x)dz = C_x(q,x).$$

Application 6: Choice of Quality

- Suppose that costs take a simple form: $C(q,x) = c(x)q$.
- The monopolist's choice of x satisfies
$$P_x(q,x) = c'(x).$$
Hence, x is chosen with regards to its influence on the price the marginal buyer will pay.
  - Thus, the marginal effect of x on the marginal buyer is key
- The socially efficient choice considers the average:
$$\frac{1}{q}\int_0^q P_x(z,x)dz = c'(x).$$
  - Thus, the marginal effect of x on the average buyer is key.

Application 6: Choice of Quality

- Summary: To quote from a famous PhD textbook on Industrial Organization:

  - “The incentive to provide quality is related to the marginal willingness to pay for quality -- for the marginal consumer in the case of the monopolist and for the average consumer in the case of a social planner.” (Tirole, 1989, page 101.)

- Depending on the circumstances, the monopolist may provide too little or too much x relative to the efficient amount.

Application 6: Choice of Quality

- **Example 1:**

  Consider the TV network’s decision to flood Olympic coverage with human interest stories that most die-hard sport fans don’t really care about. The key is that die-hard sports fans are not the marginal “buyer.” They will watch everything anyway. The “marginal” buyer enjoys the human interest stories, and so x is distorted toward that segment.

Application 6: Choice of Quality

- **Example 2: Advertising**

  x is advertising dollars. The monopolist maximizes

  $$\pi(q,x) = P(q,x)q - C(q,x).$$

  The monopolist's marginal conditions are simply

  $$P(q,x) + P_q(q,x)q = C_q(q,x),$$

  $$P_x(q,x)q = C_x(q,x).$$

  - Define the price elasticity of demand as usual:

    $$\varepsilon = \frac{p}{q}\frac{1}{P_q(q,x)},$$

    and the advertising elasticity of demand as

    $$\eta = \frac{\%\Delta q}{\%\Delta x} = -\frac{x}{q}\frac{P_x(q,x)}{P_q(q,x)}.$$

Application 6: Choice of Quality

- **Example 2**, continued:

  Using the elasticities, we rewrite the margin condition for x:

  $$P_x(q,x)q = 1 \implies \frac{\eta}{|\varepsilon|} = \frac{x}{pq}.$$

- This is the Dorfman-Steiner advertising condition.

  It says that the ratio of the price and advertising elasticities of demand should equal the ratio of advertising expenditures to total revenue.

- Empirical work supporting this condition is discussed in Cabral, ch.13.

Application 6: Choice of Quality

- **Example 3: Providing durability.**
  - It is often argued that monopolists provide poor durability in their products. While this may be true, an economist named Swan established a broad set of conditions for which this is decidedly false.
  - One should not presume that market power leads to inefficiently low durability.
- Suppose that x denotes how long a product lasts and thus consumers care only about $y=q \cdot x$. We can therefore replace the inverse demand function, $P(q,x)$, with the simpler $P(q \cdot x)$ or $P(y)$.
- Suppose (and here is an important condition for Swan's result) that $C(q,x)$ is multiplicatively separable: $C(q,x)=c(x)q$.

Application 6: Choice of Quality

- **Example 3, continued:**

  The cheapest way of producing $y=q \cdot x$ units of consumption is to choose x to minimize $c(x)/x$ (the average cost of durability).
  - This cost-minimizing x^* will satisfy $c'(x^*) = c(x^*)/x^*$. (Why?)
  - To see this note

$$\min_{x,q,\ xq=y} c(x)q = \min_{x} \frac{c(x)}{x} y.$$

  - The minimum cost of producing y is therefore $(c(x^*)/x^*)y$.
- A social planner would always choose x so as to minimize the cost of producing y.
- A monopolist will do so as well. (This is Swan's remarkable result.)

Application 6: Choice of Quality

- **Example 3, continued:**
  - Notice that

    $\pi(q,x) = P(qx)qx - c(x)q \rightarrow \pi(y,x) = P(y)y - (c(x)/x)y.$
  - The two marginal conditions can be written in terms of y as

    $P'(y)y + P(y) = c(x)/x,$

    $P'(y)y + P(y) = c'(x).$
- Thus, $c(x)/x = c'(x)$, and so $x=x^*$ is efficient.
- **Intuition:** If the consumer cares only about y, then the monopolist should provide y as cheaply as possible. This implies that durability, x, is set at the efficient level.
- Output is still too low (the standard monopoly problem), but durability is efficient.

Application 7: "Monopoly" with Competitive Fringe

- Suppose that a large number of firms behave as price takers and their supply behavior is characterized by $S_f(p)$
- One dominant firm acts as a "monopolist" on the residual demand and chooses, q_{DF}
- Price is determined by the market demand curve:

$$p = P(q_{DF} + S_f(p)) .$$

- If we solve this expression for p on the lefthand side and q_{DF} on the righthand side, we call the equation the "residual demand curve"
- Once the dominant firm has computed the residual demand curve, the firm behaves as a monopolist would with that demand relationship.

Application 7: Monopoly with Competitive Fringe

- **Example 1**: Suppose

$$p = 1000 - 4Q,\ \ C(q_{DF}) = 80q_{DF},\ \ S_f(p) = 50$$

- Because total output is $Q = q_{DF} + S_f(p)$ we can substitute into the demand equation to obtain …

$$p = 1000 - 4(q_{DF} + 50)$$

which simplifies to the residual demand curve

$$p = 800 - 4q_{DF}.$$

- The dominant firm acts as a monopolist with this demand. First, we set *MR=MC,*

$$MR(q_{DF}) = 800 - 8q_{DF} = MC = 80,$$

and thus the optimal output is $q_{DF}^{*} = 90.$ Inserting this into the residual demand curve, we have

454

$$p^{*} = 800 - 4(90) = 440.$$

Application 7: Monopoly with Competitive Fringe

- **Example 2**: Suppose

$$p = 10 - Q,\ \ C(q_{DF}) = 2q_{DF},\ \ S_f(p) = p\,.$$

- Because total output is $Q = q_{DF} + S_f(p)$ we can substitute into the demand equation to obtain …

$$p = 10 - (q_{DF} + p)$$

which simplifies to the residual demand curve

$$p = 5 - \tfrac{1}{2}q_{DF}.$$

- The dominant firm acts as a monopolist with this demand. First, we set *MR=MC,*

$$MR(q_{DF}) = 5 - q_{DF} = MC = 2,$$

and thus the optimal output is $q_{DF}^{*} = 3.$ Inserting this into the residual demand curve, we have

455

$$p^{*} = 5 - \tfrac{1}{2}(3) = 3.5$$

Oligopoly games

- Topic 1: Quantity (Cournot) games
- Topic 2: Price (Bertrand) games

Topic 1: Output (Cournot) games

- Suppose that there are *n* firms competing in a market, but *n* is sufficiently small number that no one is a price taker.
- We need to make an assumption about how the firms compete. Let's suppose that each firm i chooses an amount of output to sell in the market, q_i, and that the market price is such that demand equals to sum of the *n* firms' supply.

$$p^* = P(Q^*), \text{ and } Q^* = \sum_{i=1}^{n} q_i^*.$$

- This is sometimes called an **output game** (because firms choose their outputs and the "market" chooses the price) or a **Cournot game** (after Cournot, who first explored this in 1838).

An Aside about Game Theory

- Game Theory refers to a very structured way to model strategic interactions. For our purposes, the tools of game thoery can be used to define and find equilibria in settings like oligopoly.
- A game is defined by three things: the **players** in the game, the **strategies** each player may choose, and the **payoffs** each player gets as a function of the strategies that everyone has chosen.
- When games are very simple (e.g., 2 players, each with just a few strategies) we can present all of the relevant information in a payoff matrix.

Games with Payoff Matrices

- **Prisoner's dilemma**. Two players, A and B.

	Confess	Don't confess
Confess	-8, -8	0, -10
Don't confess	-10, 0	-1, -1

Game theory prediction: both prisoners choose "Confess"

- Variation with firms choosing whether or not to expand output:

	Don't Expand	Expand
Don't Expand	18, 18	15, 20
Expand	20, 15	16, 16

Game theory prediction: both firms "Expand"

-

Coordination Games

- Suppose that players A and B agree to meet at noon in New York City, but before they can agree on the location, their telephone call gets cut off. Both imagine there are only two reasonable places they would have agreed to -- the Information Desk at Grand Central Station, or the corner of Times Square.

	Times Square	Grand Central Station
Times Square	2, 2	0,0
Grand Central Station	0,0	1,1

- Now what is a reasonable prediction? Game theory actually provides 3, although one is "focal".
- Suppose instead that the payoff to Times Square is (3,2) and the payoff to Grand Central Station is (2,3). Now what?

Games with lots of strategies

- When each player has access to many strategies, such as choosing any output level from 0 to 1000, the payoff matrix is not convenient. Instead, we rely on functions and mathematics.
- The Cournot output game is an example of such a setting. We can write firm A's payoff as a function of firm A's output, q_a , and a function of B's output, q_b . In this case with two firms, we have

$$\pi_A(q_a, q_b) = P(q_a + q_b)q_a - C_A(q_a),$$
$$\pi_B(q_a, q_b) = P(q_a + q_b)q_b - C_B(q_b).$$

- How should we expect the firms to choose their output?

Nash Equilibrium

- Whether we write the game as a payoff matrix or using profit functions, the basic notion of **equilibrium** is the same and is due to John Nash. A **Nash equilibrium** is a set of strategies, one for each player, such that each player's choice is optimal given the choices of the others.
- Nash received the Nobel prize for both developing the definition of Nash equilibrium, and for showing that one always exists in a large class of well-behaved games.
- In our first two matrix games, the Nash equilibrium is easy to find. Indeed, choosing "Confess" in the first game and "Expand" in the second game is optimal, regardless of what the other player is doing. When this is the case, the equilibrium has the further property that the players' **strategies** are **dominant**.
- For Cournot games, we will use marginal analysis to find the Nash equilibrium.
-

Back to the Cournot Output Game

- Suppose that (q_a^*, q_b^*) is a Nash equilibrium. Then it must be that if firm A correctly predicts firm B will choose q_b^*, that firm A will want to choose q_a^*:

$$\pi_A(q_a^*, q_b^*) = P(q_a^* + q_b^*)q_a^* - C_A(q_a^*) \geq \pi_A(q_a, q_b^*) = P(q_a + q_b^*)q_a - C_A(q_a),$$

  or, in terms of marginal analysis,

$$\frac{\partial \pi_A(q_a^*, q_b^*)}{\partial q_a} = P(q_a^* + q_b^*) + P'(q_a^* + q_b^*)q_a^* - C_A'(q_a^*) = 0.$$

  But this equation is just requiring that firm A's marginal revenue (using the residual demand curve given B's output) is set equal to marginal cost.
- It is ***as if*** each firm behaves like a monopolist with residual demand.

Topic 1: Output (Cournot) games

- Suppose that there are 2 firms, each with costs, *C(q)=2q.*
- Suppose that the market demand curve is

$$P(q_a + q_b) = 20 - (q_a + q_b).$$

- What is the Nash equilibrium in this output game?
  - Step 1: For a given output from firm B, what is firm A's residual demand curve? What is firm A's marginal revenue using this curve?

$$P_a(q_a) = (20 - q_b) - q_a,$$
$$MR_a(q_a) = (20 - q_b) - 2q_a.$$

Topic 1: Output (Cournot) games

- Step 2: What is firm A's best response given an output conjecture for B?

$$\begin{aligned} MR_a(q_a) &= MC_a(q_a) \\ (20 - q_b) - 2q_a &= 2. \end{aligned}$$

  Solving for A's optimal output as a function of firm B's output, we have the following relationship, called firm A's **reaction function**:

$$q_a = R_a(q_b) = 9 - \frac{1}{2}q_b.$$

  For example, if firm A thought firm B would supply 4 units, firm A's optimal supply is 7 units.

- Notice that firm A's reaction function indicates that A will supply less in response to a belief that firm B will supply more.

Topic 1: Output (Cournot) games

- Step 3: Perform the same steps for the other firm. Because firm A and B have the same cost functions, we will get a symmetric reaction function for B:

$$q_b = R_b(q_a) = 9 - \frac{1}{2}q_a.$$

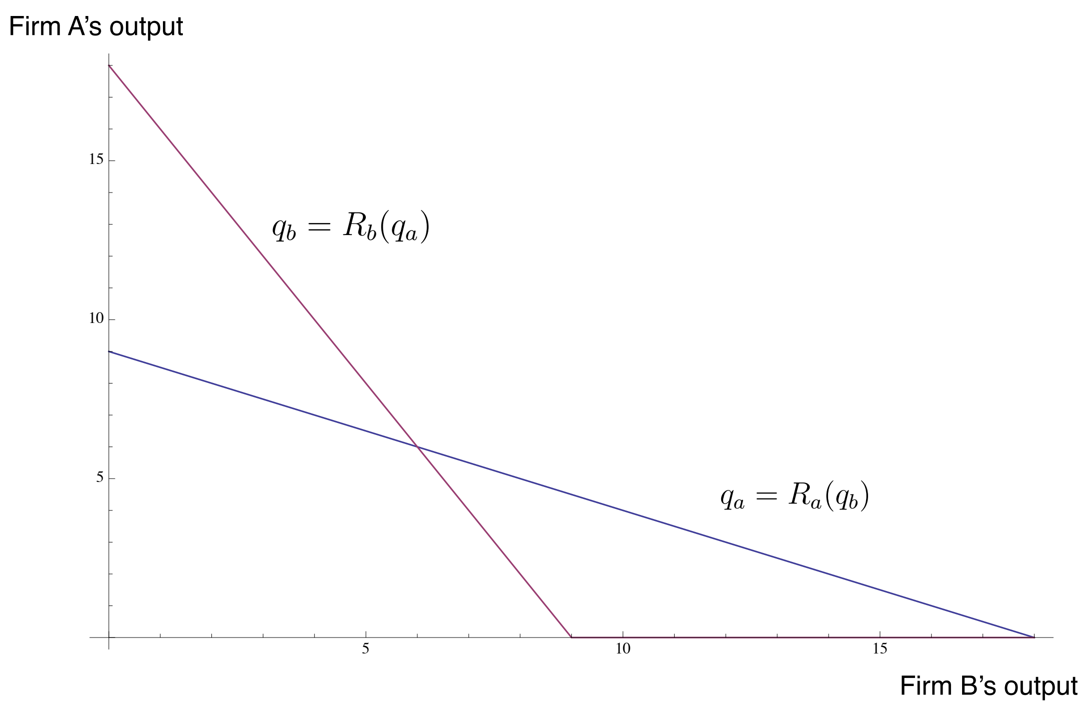
- Step 4: Find the intersection of the reaction functions. The point where they intersect with be the point where the outputs are jointly optimal given each other. This is the Nash equilibrium, of course.

Using algebra, we can solve for the intersection of the reaction functions. It is simple to verify that

$$q_a^* = q_b^* = 6$$

is the solution for this problem.

Topic 1: Output (Cournot) games



Topic 1: Output (Cournot) games

- Let's return to our **taxicab setting** with one high-capacity firm and 5 low-capacity firms: recall we have 5 firms with capacity of 10,000 miles, and 1 high-capacity firm with capacity of 20,000 miles.
- **Claim**: In equilibrium, the 5 low-capacity firms produce q_L^*=10,000 miles and the high-capacity firm produces q_H^*=15,000 miles.
- First, consider the high-capacity firm's optimal output given that the other 5 firms are choosing $Q_{-H}^* = 5q_H^* = 50{,}000$ in total.
  - The high-capacity firm's marginal revenue is
    $$\begin{aligned} MR_H(q_H) &= P(q_H + 50{,}000) + P'(q_H + 50{,}000)\, q_H \\ &= 1 - (q_H + 50{,}000)/100{,}000 - q_H/100{,}000 \\ &= 0.50 - q_H/50{,}000. \end{aligned}$$
  - At q_H^*= 15,000 we have $MR_H(15{,}000) = 0.20 = MC$.
  - Thus, q_H^*= 15,000 is optimal providing $Q_{-H}^* = 50{,}000$.

Topic 1: Output (Cournot) games

- Second, consider the decision of a low-capacity firm given that q_H^*=15,000 and the other low-capacity firms each choose q_L^*= 10,000. In this case, $Q_{-L}^* = 4\, q_L^* + q_H^* = 55{,}000$.
  - The low-capacity firm's marginal revenue given $Q_{-L}^* = 55{,}000$ is
    $$\begin{aligned} MR_L(q_L) &= P(q_L + 55{,}000) + P'(q_L + 55{,}000)\, q_L \\ &= 1 - (q_L + 55{,}000)/100{,}000 - q_L/100{,}000 \\ &= 0.45 - q_L/50{,}000. \end{aligned}$$
  - At q_L^*= 10,000 we have $MR_L(10{,}000) = 0.25 > MC$.
  - The low-capacity firm would like to choose more output, but is capacity constrained, so q_L^*= 10,000 is optimal.
- Hence, ***q*$_H$*= 15,000 and *q*$_L$*= 10,000 is a Nash equilibrium**. The price is p^*= 0.35, which is greater than the competitive price of 0.20, but less than the monopoly price of 0.60.

Topic 1: Output (Cournot) games

- We can generalize the previous game with n firms and more general demand and cost functions. The ideas are the same
  - (1) the firms should correctly anticipate the market output
  - (2) each firm must maximize its profits given the outputs of the others.
- Suppose that firm i anticipates that the other firms will choose (in total) to sell Q_{-i}^* output. If firm i is correct and brings q_i to market, the market price will be

$$p = P(Q_{-i}^* + q_i).$$

- Thus, from firm i's point of view, it is as if firm i is a monopolist with a residual demand curve equal to

$$P_i(q_i) = \mathrm{P}(Q_{-i}^* + q_i).$$

- Applying the marginal output rule from monopoly, we have

$$MR_i(q_i) = P(Q_{-i}^*+q_i) + P'(Q_{-i}^* + q_i)\, q_i = MC_i(q_i).$$

Topic 1: Output (Cournot) games

- In equilibrium, each firm i=1,…,n chooses q_i to satisfy

$$MR_i(q_i) = P(Q_{-i}^*+q_i) + P'(Q_{-i}^* + q_i)\, q_i = MC_i(q_i).$$

- We have n equations and n unknown quantities. Under mild conditions, there will be a unique set of equilibrium outputs.
- The key insight in the Cournot output game is that we can model oligopoly as n interacting monopolies, each with a downward-sloping residual demand curve. Thus, the intuition of monopoly extends to more interesting settings.
- Generally, with homogenous goods one can show that …
  - the oligopoly market price will lie between the competitive price and the monopoly price and,
  - that industry profit will be positive, but not as great as the monopoly level.

Remarks on the Cournot Game

- **To summarize**, the previous analysis easily generalizes to any number of firms (i.e., $n>2$), using general cost and demand functions. Remarkably, the inverse-elasticity rule has a very simple form as a function of the number of firms, *n,* when all firms have symmetric cost functions:

$$\frac{P(Q^*) - MC(q^*)}{P(Q^*)} = \frac{1}{n|\varepsilon^d|}$$

  The interesting feature to note is that if $n=1$, we recover the monopoly formula. If n is close to infinity, the market price is close to marginal cost (i.e., the perfect competition outcome). The **Cournot model** therefore has the nice feature that it **includes perfect competition and monopoly as special cases**.
- All of the Cournot analysis only makes sense if (1) goods are homogenous and (2) firms choose outputs and the market determines the equilibrium price.

Day 9:
Oligopoly, capturing surplus (price discrimination, screening, market segmentation)

Topic 2: Price Games

- The **Cournot/Output games** we have considered have two key features:
  - 1) The good is homogenous (e.g., commodity markets)
  - 2) The firms choose outputs and the "market" determines price
- **Note:** Output games nicely model situations in which firms simultaneously choose their productive capacities and then effectively dump all of their output on the market. In this sense, the Cournot game is akin

- In **price games (a.k.a. Bertrand games)**, the firms choose prices and commit to supplying whatever is demanded at those prices.

  Note: This is not a good description of the short-run strategic setting for firms that are producing near their capacities.

Topic 2: Price games, continued

- We can further distinguish price games according to whether the goods are **perfect substitutes** or **differentiated products.**
- **Betrand games with perfect substitutes.** When the goods are perfect substitutes and firms have identical marginal costs, the Nash equilibrium in prices is rather dramatic: ***p*=MC****.*
  - Compare this to the Cournot game and its equilibrium. There, firms received a market price above MC and made profits.
  - Fundamentally, competing over prices with unlimited capacity is a very competitive situation compared to Cournot where firms compete by choosing capacities. This is the key insight of **Bertrand v. Cournot.**
- **Betrand games with differentiated products.** Once goods are no longer perfect substitutes, undercutting a rival is less valuable. With perfect substitutes, undercutting by one penny results in 100 percent market share. With differentiated products, a slight reduction in price leads to a slight increase in demand.

Topic 2: Price games, continued

- **Differentiated Bertrand games.** Suppose that there are two firms, A and B, each selling a demand substitute. The relationship between prices and demands are given by ...

$$q_a = D_a(p_a, p_b),$$

$$q_b = D_b(p_a, p_b).$$

For concreteness, let's suppose that

$$D_a(p_a, p_b) = 14 - 2p_a + p_b,$$

$$D_b(p_a, p_b) = 14 - 2p_b + p_a.$$

Also suppose that each firm has constant unit cost, *AC=MC=c*. Firm A's profit as a function of its price and firm B's price is thus

$$\pi_a(p_a, p_b) = (p_a - c)(14 - 2p_a + p_b).$$

Topic 2: Price games, continued

$$\pi_a(p_a, p_b) = (p_a - c)(14 - 2p_a + p_b).$$

- As before, we can solve for firm A's optimal price as a **reaction function** which depends upon firm B's price. We apply a form of the marginal output rule -- this time determining margins with respect to price (rather than quantity). Thus, we have the condition

$$\begin{aligned} \frac{\partial \pi_a(p_a, p_b)}{\partial p_a} &= (14 - 2p_a + p_b) - 2(p_a - c) \\ &= 14 + 2c - 4p_a + p_b = 0. \end{aligned}$$

- Suppose *c=2*. Solving for firm A's reaction function, we obtain

$$p_a = R_a(p_b) = \frac{9}{2} + \frac{1}{4}p_b.$$

Topic 2: Price games, continued

$$p_a = R_a(p_b) = \frac{9}{2} + \frac{1}{4}p_b.$$

- Performing the same strategic analysis for firm B, we obtain

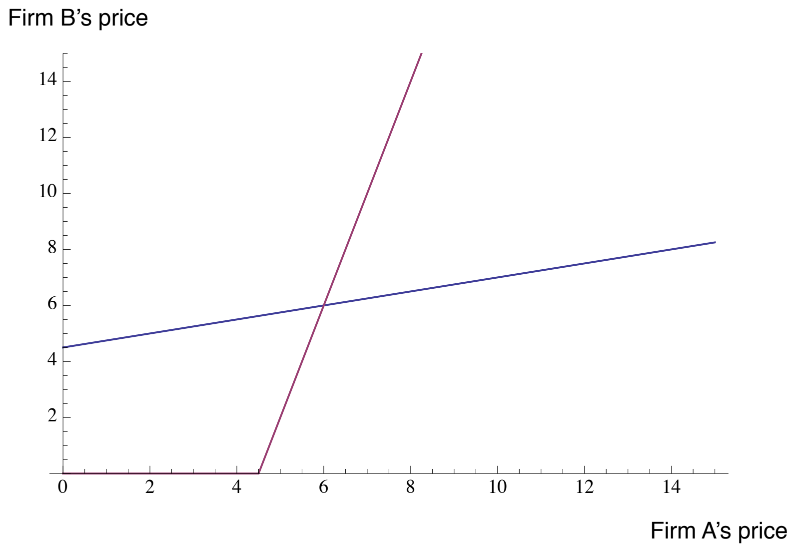
$$p_b = R_b(p_a) = \frac{9}{2} + \frac{1}{4}p_a.$$

- Note that these reaction functions are upward sloping, in contrast to the reaction functions in the Cournot game. E.g., a higher price by a rival induces a firm to raise their own price.

Topic 2: Price games, continued

- The Nash equilibrium in this game is found to be (after some maths)

$$p_a^* = p_b^* = 6.$$



- Unlike the case of perfect substitutes, each firm's equilibrium price exceeds its unit cost and firms make economic profit.

Capturing consumer surplus

- Topic 1: A surplus-capturing benchmark for monopoly
- Topic 2: Direct price discrimination
- Topic 3: Indirect price discrimination

Topic 1: A surplus-capturing benchmark

- Recall the source of the inefficiency that plagues monopoly:

$$MR(q) = P(q) + P'(q)\,q$$

(this term is critical)

- If the monopolist could (1) identify each consumer's marginal willingness to pay for each unit on consumption, and (2) if the firm could charge a unit price specialized to each unit of each consumer's purchase, then the second term disappears:

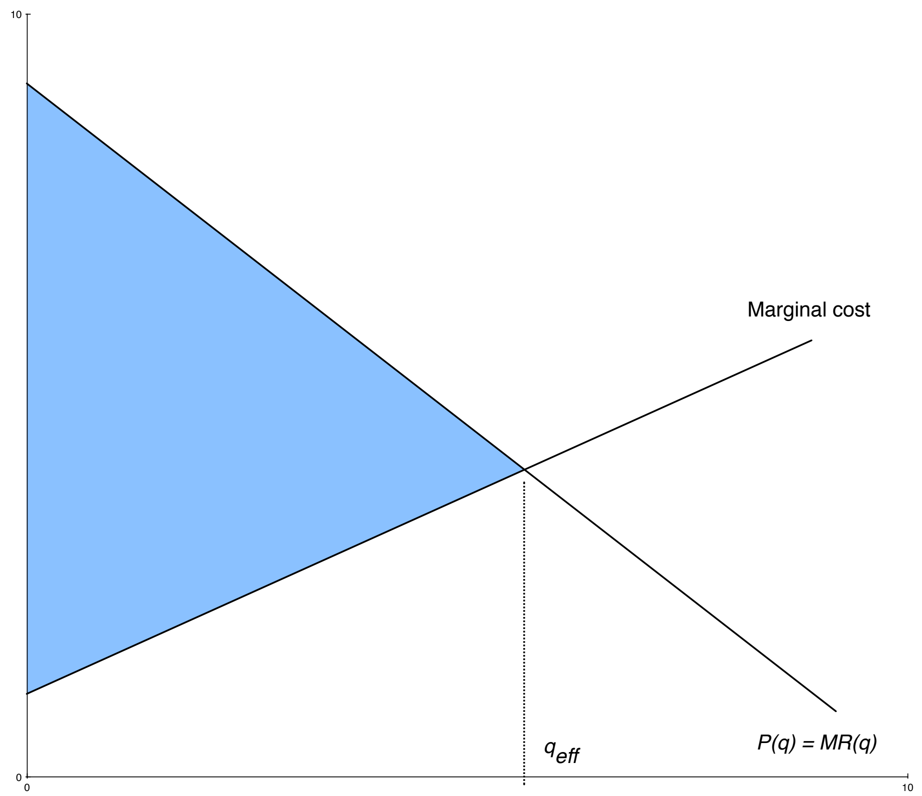
$$MR(q) = P(q) + \cancel{P'(q)\,q}\,.$$

- In this case, we say the firm can perfectly price discriminate and implements first-degree price discrimination.

Topic 1: A surplus-capturing benchmark

- The case of perfect price discrimination is a useful benchmark with several immediate properties:
  - Marginal output rule implies $P(q^*)=MC(q^*)$. Thus, the monopolist chooses the socially efficient level of output.
  - No surplus is left to consumers. The aggregate surplus goes entirely to monopoly profits.
  - No deadweight loss arises. Intuitively, if the monopolist captures all of the consumer surplus, there is no reason to price above marginal cost.
- A uniform-pricing monopolist faces a tradeoff between taking a "bigger slice of the pie" against making the "pie smaller." A perfect-price-discriminating monopolist gets the whole pie.

Perfect (first-degree) price discrimination



Topic 1: A surplus-capturing benchmark

- Perfect price discrimination is mainly a benchmark, not a practical pricing policy.
  - In general, firms are not able to identify all of the relevant variations in consumers' willingness to pay, and firms are unable to prevent consumers from trading among themselves. This limits what can be done in practice.
- Ingredients for price discrimination: To practice any form of price discrimination, three conditions must be satisfied:
  - the firm must have **market power** (e.g., monopoly)
  - the firm must be able to **identify** the relevant consumer **segments** (either directly or indirectly)
  - the firm must be able to **prevent arbitrage** between consumers facing different prices

Topic 1: A surplus-capturing benchmark

- Taxonomies of price discrimination.

  Old school. Pigou wrote a popular textbook in the 1930s which categorized price discrimination in three degrees.
    - First-degree price discrimination: all of the consumer surplus is captured and converted to revenue
    - Second-degree price discrimination: an approximation to first-degree price discrimination (discussed below)
    - Third-degree price discrimination: when the monopolist is able to separate its consumers into distinct markets, but within each market sets a single uniform price.
  - This taxonomy is bad for a few reasons. It is incomplete and Pigou's discussion of second-degree price discrimination doesn't make a lot of sense. Indirect PD?

Topic 1: A surplus-capturing benchmark

- **Modern taxonomy:** Price discrimination is **direct** or **indirect**.

  Direct price discrimination: the firm offers prices based upon directly observable consumer characteristics.

  - Third-degree price discrimination is an example, but direct price discrimination doesn't necessarily require uniform pricing. (E.g., each identifiable segment may be offered different 2-part tariffs - e.g., Disneyland family passes, etc.)
  - geography (country, zip code, etc.)
  - time of day
  - purchase history
  - demographics (student, AARP card, family, etc)

Topic 1: A surplus-capturing benchmark

- **Modern taxonomy:** Price discrimination is **direct** or **indirect**.

  Indirect price discrimination: the firm offers all consumers the same "menu" of products and prices, but the menu is designed so that consumers with different valuations will select different options from the menu.

  - quantity discounts
  - two-part tariffs (laser printers and toner, razors and razor blades, IBM mainframes and punchcards, Xerox copiers and paper, admission fees and usage fees, etc.)
  - quality variations (vertically differentiated product line); "damaging" goods to segment the market
  - coupons, rebates
  - bundling discounts

Topic 1: A surplus-capturing benchmark

- **Pigou's taxonomy** versus **modern taxonomy**:
  - Everyone equates Pigou's first-degree price discrimination with the idea that the seller extracts all consumer surplus, i.e., perfect price discrimination.
  - A charitable interpretation of Pigou's second-degree price discrimination is that he really meant indirect price discrimination. Indeed, most economists refer to the "indirect" price discrimination strategies (e.g., quantity discounts, etc.) as 2nd-degree price discrimination, even though Pigou didn't have these ideas in mind.
  - Third-degree price discrimination is the predominant form of direct price discrimination. Sometimes economists use the phrase "third-degree" to mean any form of direct price discrimination. This is broader than Pigou's definition.
- When talking to economists and marketers, "3rd degree" and "direct" price discrimination are roughly synonymous; likewise, "2nd degree" and "indirect" are roughly synonymous.

Topic 2: Direct price discrimination

- **Direct price discrimination** is the practice of identifying separate groups of buyers using some observable characteristic and charging different prices to each group.
- The simplest example of direct price discrimination is the case in which a monopolists can identify multiple market segments and can set a distinct uniform price for each segment. (i.e., an example of Pigou's third-degree price discrimination).
- Example: Consider the case of two discernible types of consumers with different demand elasticities: senior citizens and non-seniors.
  - Suppose that the senior segment has a demand curve

$$p_s = P_s(q_s) = 150 - q_s.$$

  - Suppose that non-senior segment has a demand curve

$$p_n = P_n(q_n) = 400 - 2q_n.$$

Topic 2: Direct price discrimination

- The marginal output rule can be applied to each segment:

$$MR_s(q_s) = 150 - 2\,q_s = MC_s(q_s),$$

$$MR_n(q_n) = 400 - 4\,q_n = MC_n(q_n).$$

- Suppose that each segment had the same constant marginal cost, $MC{=}c$. Then we could solve for the monopoly output in each segment in isolation.
- If instead the marginal cost in one segment depends upon the output in the other, the marginal-output conditions will need to be solved simultaneously. For example, suppose the cost function is $C(q_s, q_n){=}(1/2)(q_s{+}q_n)^2$. Then

$$150 - 2\,q_s = q_s{+}q_n,$$

$$400 - 4\,q_n = q_s{+}q_n.$$

Which has the solution $q_s^*{=}25$, $q_n^*{=}75$. The associated prices in each segment are therefore $p_s^*{=}150{-}q_s^*{=}125$ and $p_n^*{=}400{-}2q_n^*{=}250$.

Topic 2: Direct price discrimination

- It is certainly possible that the marginal costs across segments are not equal.
  - It may be that senior citizens require more customer service, in our example.
  - If the segments are determined geographically, it may be that the marginal costs of delivering the product differ across segments.
- When the marginal costs are the same, however, it follows that the marginal revenues across the segments must also be equal (assuming the monopolist sells in both segments). E.g.,

$$MR_1(q_1) = MC(q_1{+}q_2),$$

$$MR_2(q_2) = MC(q_1{+}q_2),$$

and so $MR_1(q_1){=}MR_2(q_2)$.

Topic 2: Direct price discrimination

- Using the elasticity formula for marginal revenue and the result that $MR_1(q_1^*) = MR_2(q_2^*)$, we have

$$MR_1(q_1) = p_1\left(1 + \frac{1}{\varepsilon_1}\right) = MR_2(q_2) = p_2\left(1 + \frac{1}{\varepsilon_2}\right),$$

  or more simply, we have the price ratio entirely determined by

$$\frac{p_1}{p_2} = \frac{\varepsilon_1(1+\varepsilon_2)}{\varepsilon_2(1+\varepsilon_1)}$$

- Again, remember that this is a result of assuming (1) that marginal costs are the same for each segment and (2) that each segment is supplied.

Topic 3: Indirect price discrimination

- **Indirect price discrimination** with nonlinear pricing.

  Before we get into the details of nonlinear pricing, consider a simple setting in which there are three types of consumers (in equal proportion) and their demand data are given in the following table:

unit	Type A	Type B	Type C
1st	7	9	11
2nd	5	7	9
3rd	3	5	7
4th	1	3	5
5th	0	1	3
6th	0	0	1

- Notice that each consumer is willing to purchase at most 6 units.
- In what follows, we assume that marginal and average costs are zero. If not, we will need to incorporate units costs.

Topic 3: Indirect price discrimination

- Just to be clear we understand the table, ...

unit	Type A	Type B	Type C
1st	7	9	11
2nd	5	7	9
3rd	3	5	7
4th	1	3	5
5th	0	1	3
6th	0	0	1

  - Suppose that the monopolist can perfectly price discriminate; then how much revenue can be obtained? R=16+25+36=77.
  - If the monopolist can only set uniform price, what is the optimal price to set? What is the revenue? Answer: $p^*=5$, $R=45$.
  - If the monopolist can directly price discriminate by type, but must set a uniform price for each type, what are the optimal prices? Answer: $p_a^*=5$, $p_b^*=5$, $p_c^*=7$, $R=46$.

Topic 3: Indirect price discrimination

- Suppose now that the monopolist cannot observe the consumer's type, but can observe which unit the consumer is purchasing. The firm can therefore set six distinct prices, one for the first unit, p_1, one for the second unit, p_2, and so on. What are the optimal prices?

unit	Type 1	Type 2	Type 3	best marginal price	number sold	revenue
1st	7	9	11	7	3	21
2nd	5	7	9	5	3	15
3rd	3	5	7	5	2	10
4th	1	3	5	3	2	6
5th	0	1	3	3	1	3
6th	0	0	1	1	1	1
Total						56

- Notice that revenues of R=56 can be achieved with the prices $p_1^*=7$, $p_2^*=5$, $p_3^*=5$, $p_4^*=3$, $p_5^*=3$, $p_6^*=1$.

Topic 3: Indirect price discrimination

- Revenues of $R=56$ are achieved with prices

$$p_1^*=7,\ p_2^*=5,\ p_3^*=5,\ p_4^*=3,\ p_5^*=3,\ p_6^*=1.$$

- It is also possible to offer only three packages and obtain the same outcome: the first two units are sold in a package for $p_{\{12\}}^*=12$, the next two units are sold in a package for $p_{\{34\}}^*=8$, and the last two units are sold in tandem for $p_{\{56\}}^*=4$.
  - Aside: as a practical matter, a firm can often get very close to the revenues of the optimal nonlinear pricing scheme by using just a few prices and packages.
- Note that in the example above, that the optimal unit prices are decreasing (quantity discounts). The average prices are also falling:

average price of 2 units = 6,

average price of 4 units = 5,

average price of 6 units = 4.

Topic 3: Indirect price discrimination

- Even when multiple prices are not available, often times a simple two-part tariff can do almost as well.
- A two-part tariff is a price that has a fixed component and a marginal component:

$$T(q) = F + pq.$$

- Examples:
  - amusement parks (admission fee and per-ride fee)
  - clubs (membership fee and marginal usage fees)
  - laser printers, razors, etc. (e.g., base product and marginal supplies)
  - IBM and punchcards (1930s); Xerox copiers and paper.

Topic 3: Indirect price discrimination

- Returning to our example, notice that the following 2-part tariff does a pretty good job at getting most of the surplus:

  $$T(q) = 6 + 3q.$$

  - Given the marginal price of p=3, a type A consumer obtain gross consumer surplus of

    $$CS_a(p)= CS_a(3)=(7\text{-}3)+(5\text{-}3)+(3\text{-}3)=6.$$

    Thus, type A will be just willing to participate.
  - It is straightforward to check that types B and C will strictly prefer to participate if type A is willing to do so.
  - The total revenue earned by this 2-part tariff is therefore

    $$(6+3(3)) + (6+3(4)) + (6+3(5)) = 54.$$
- The two-part tariff revenue of R=54 is close to the revenue obtained from the six prices, R=56.
- As a practical matter, computing the optimal 2-part tariff proceeds using data and numerical methods. Even our example above is hard to calculate.

Topic 3: Indirect price discrimination

- So far, we have demonstrated that nonlinear pricing is very similar to direct price discrimination, except that the segments are instead the numbered units rather than the consumer types.

- We will now see that segmenting the market via quality variations is mathematically equivalent, even though it feels very different.

Topic 3: Indirect price discrimination

- **Indirect price discrimination** using **quality variation**.

  Introducing quality distortions into a product line is a way to indirectly price discriminate.
- In 1849, Jules Dupuit studied the conditions of third-class rail travel in France and had a remarkable insight:

  “It is not because of the few thousand francs which would have to be spent to put some roof over the third-class carriages or to upholster the third-class seats that some company or other has open carriages with wooden benches; it would be a small sacrifice for popularity. What the company is trying to do is to prevent the passengers who can pay the second-class fare from traveling third class; it hits the poor not because it wants to hurt them, but to frighten the rich.”
- Does this sound like modern-day air travel with required Saturday-night stayover requirements, minimal legroom, bad food?

Topic 3: Indirect price discrimination

- Let's consider the example of air travel.
  - Suppose that an airline has two tickets to sell: a high-quality "good" ticket and a low-quality "bad" ticket. Both tickets get you to your destination, but the high-quality good ticket is always preferred to the low-quality ticket.
  - Suppose that there are two groups of consumers: business customers and vacationers. The airline cannot directly distinguish the two groups, but (1) business customers are always willing to pay more for either ticket than vacationers, and (2) business customers have a larger value differential between the high- and low-quality tickets.
    - (2) is ofter referred to as the "sorting condition" or the "single-crossing property"

Topic 3: Indirect price discrimination

- As an example of (1) business customers willing to pay more for either ticket than vacationers, and (2) business customers have a larger value differential between the high- and low-quality tickets, let's use these numbers:
  - The business travelers value a good ticket at 800 and a bad ticket at 400.
  - The vacationing travelers value a good ticket at 300 and a bad ticket at 250.
- If the airline could observe who was a business customer and who was a vacationer, then they would only sell good tickets and charge the business traveler 800 and the vacationer 300. This would earn \$550 in average revenues per customer.
- If the airline is only able to sell good tickets at a uniform price, the firm would choose to sell only to business customers at p=800 and thus earn average revenues per-customer of \$400.

Topic 3: Indirect price discrimination

- Assuming that the airline can not observe the type of customer who is purchasing the ticket, how can it improve its revenues?
- Let's return to our methods for quantity discounts to see the connection with value pricing. Note that the values for the various tickets can be re-arranged into this table:

unit	Business	Vacationer	best marginal price	# sold	revenue
base quality – ticket with stayover	400	250	250	2	500
added quality – removal of stayover	400	50	400	1	400
Total					900

- It follows that the optimal price for the "bad" ticket (first unit) is p_{bad}=250 and the charge the for the second unit (the upgrade to higher quality) is $p_{upgrade}$=400. Hence, the good ticket sells for the sum of the two components: p_{good}=250+400=650. Average revenues per consumer are now $450.

Topic 3: Indirect price discrimination

- The previous analysis can easily be applied when the proportions of the two customer groups is not 50-50 or when marginal costs are not zero. Convince yourself that you understand how things generalize.
  - If we change the analysis to the case where the business travelers outnumbered the vacationers by 2 to 1, it is best to charge p_{bad} = 400 and sell to 2/3's of the market rather than set p_{bad} = 250 and sell to everyone. Thus, the bad ticket is never purchased by itself; only the good ticket is sold.
  - If the vacationers outnumbered the business travelers 9 to 1, we would have p_1=250 and p_2=50. The latter arises because getting p_2=50 from the entire market is better than getting p_2=250 from 10 percent of the market. Thus, only good tickets are sold and p_{good}=300.
  - The quantity-discounting approach is quite flexible. Fundamentally, this strategy requires that you think about your goods as containing several increments of quality, and you price each increment optimally taking into account your consumer population.

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- We sometimes observe firms damaging goods to produce quality variation.
  - The damaged good often costs more to produce since it is the original high-quality good with additional materials or labor to reduce its quality!
- Examples:
  - 486SX versus 486DX processors. The 486SX is a coprocessor disabled 486DX, which costs more to produce than the 486DX.
  - Video zooms on cameras.
  - Fedex (and other overnight delivery services).
  - Scientific and programmable calculators.
  - IBM LaserPrinter E is IBM's low cost alternative to its popular Laser Printer in 1990. The LP-E is an LP with a controller card which purposefully slows the print speed from 10ppm to 5ppm.
  - Methyl-Methacrylate (M-M) is a plastic adhesive sold for medical and non-medical uses. Rohm-Haas purposefully started a rumor that its non-medical M-M which sells for $.85/lb was tainted with arsenic so that dental firms would not use it and instead purchase the $22/lb dental-grade M-M.